



TECHNICAL DOCUMENTATION

for CDI 605T USERS

System Software Release 1.3



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PHILIPS IMS Interactive Media Systems	INTRODUCTION
CDI 605T System Software Rel 1.3 TECHNICAL MANUAL	
Version : 1.1 Chapter : 2.0 Status : Released	

Dear customer,

This is the technical documentation for the 605T series of CD-i professional players from Philips. It is intended to be valid for latest version released for each player. This documentation is especially made to be able to support business development.

Information about the startup of the system, such as the Configuration Status Descriptor and the SYSGO process, are available here. Specifications of communication ports, input devices and extensions are also added.

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PHILIPS IMS Interactive Media Systems	CONFIGURATION STATUS DESCRIPTOR
CDI 605T System Software Rel 1.3 TECHNICAL MANUAL	
Version : 1.2 Chapter : 3.0 Status : <i>Released</i>	

1 Contents

2. Introduction
3. Updating the CSD file with DSD's from CSD memory modules
4. Updating the CSD file depending on devices present

2 Introduction.

In non volatile RAM, the CSD file is stored. This file contains the actual devices. There are two modules that operate on this file during startup : csdinit and config. Csdinit is a so called "kernel customizing module" which is called during the kernel initialization. When an application is started, also the launcher updates the CSD file. A new DSD for an open system should be approved by the CCB (Change Control Board).

3 Updating the CSD file with DSD's from CSD memory modules.

The csdinit module has another task besides updating the csd file. First, it installs a routine for the F\$Panic system call which will be executed by the kernel when there are no more processes, that is when the CD-i application exits. This routine will reset the player to prevent the player of being in a deadlock situation.

Next, the customizing module csdinit will locate the CSD data modules in memory and copy all Device Status Descriptors found to the CSD file in non volatile RAM (/nvr). If this file does not yet exist, it is created by csdinit. Any DSD's in the CSD file that are not referenced (any more) by DSD's in the CSD data modules in memory are deleted from the CSD file. The DSD for entry 0 (operating system revision level) is always over written with the DSD in memory, others are not changed.

Finally, the DSD for video is updated in the CSD file depending on the video mode (PAL/NTSC).

4 Updating the CSD file depending on devices present.

Config will first determine *which devices are present* in/connected to the player. It checks for a floppy drive, pointing devices connected to port #1 and port #2 and a terminal connected to port #3. The CSD file will be updated depending on the devices found.

If a floppy drive is present, DSD's for all available floppy formats will be added to the CSD file (both RBF and PCF formats). This check is done via a controlled bus error (the chkflop program is forked, if its exit status is 'bus error', then there is no floppy drive present).

The presence of a terminal is checked via the CTS line of port #3. If a terminal is connected, then a DSD for '/term' will be added to the CSD file. This DSD will be the first terminal entry in the CSD file. There is always a DSD for a SCF device on port #4 present.

Config will check on port #2 to see if a CD-i input device is connected via the identification protocol (RTS low for at least 10 ms, then high - device sends its ID byte). Possible devices on port #2 are: mouse/trackerball, joystick, graphic tablet, touch screen, keyboard, modem.

The device identification on port #1 is done by the slave. Config will ask the slave for the type of device found. On port #1 mouse/joystick, graphic tablet, touch screen and keyboard can be connected. The slave will combine the information from the remote control joystick with the information from a connected pointing device.

The default pointing device (the first one in the CSD file) is determined via the following rules:

- when a pointing device is connected to port #1, it will be the default pointing device no matter what device is connected to port #2.
- a pointing device connected to port #2 will be the default pointing device when no pointing device is connected to port #1 at startup, otherwise it will be the secondary pointing device. The pointing device on port #1 will be in the CSD as a secondary device when the device on port #2 is the default device.
- when no pointing device is connected to either port, port #1 will be the default pointing device. (It is assumed that a remote control will be used).
- when no input device is connected to port #2, it will become a standard RS232 (SCF) device.(see rs232 port specification - chapter 8).

On port #1, it is always possible to connect a pointing device later on or to change the pointing device connected (for example: mouse instead of graphic tablet).

On port #2, the device has to be present at startup, otherwise it will not be recognized (the identification is done only once).

Remark :The update of the CSD file is done only once, connecting a pointing device to port #1 later on will not change the default pointing device.

E.g. CDI 605T

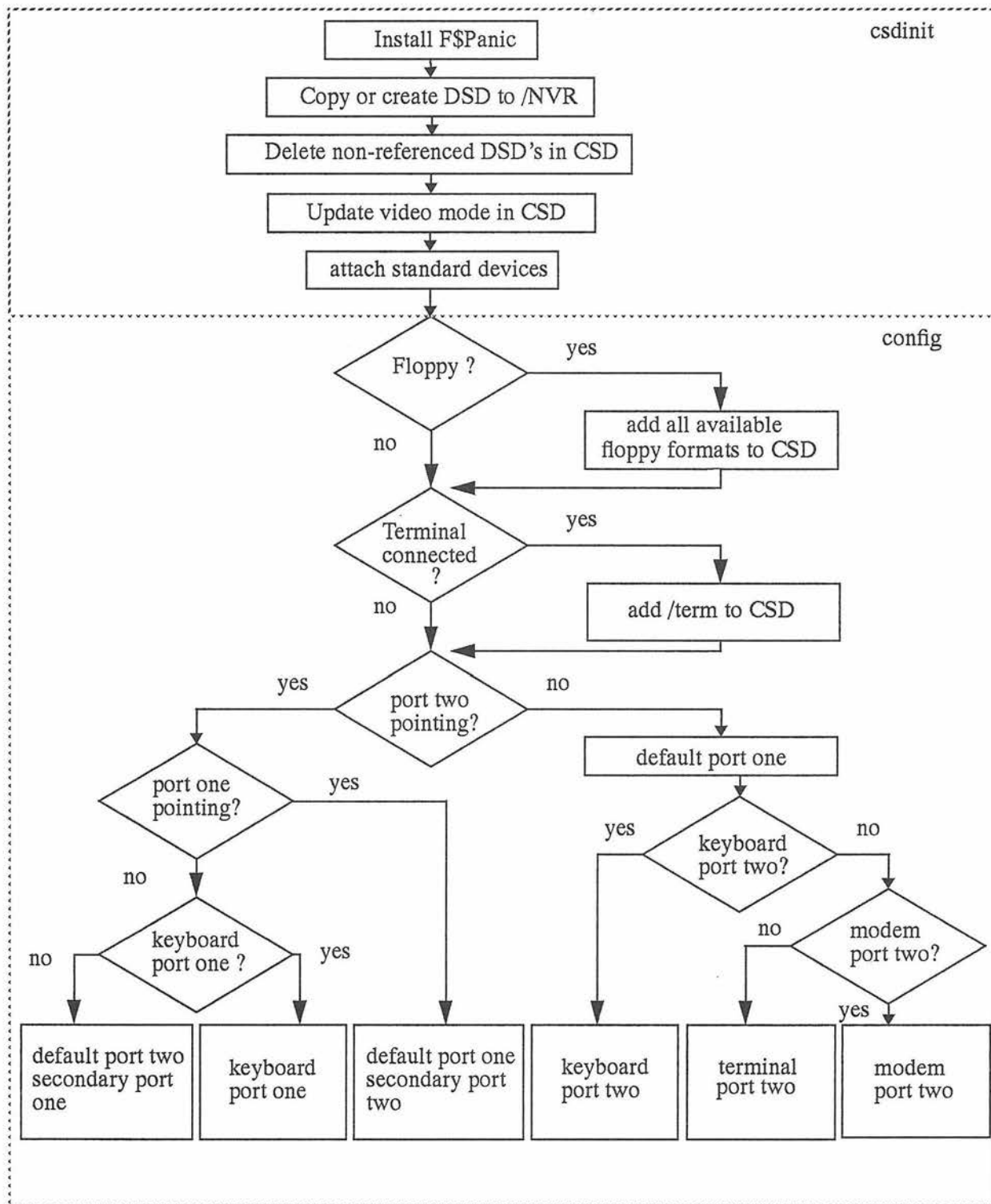
CSD file

```
0:sys:RV="1.1":
1:/cd:
2:/ap:
3:/vid2:LI="525":TV:
4:/nvr:
5:/ptr:CL="c":
6:cdp:
8:mon:
9:/pipe:
10:/kb:KG="ALL":HE="Help":ME="Menu":
11:/pck:KG="8":
20:/term:
20:/t1:
40:/d0:
40:/d0tb:
40:/d0h:
```

Updating the CSD file depending on devices present.

40:/d0hd:
40:/d0uv:
60:/pcd0:
60:/pcd0h:
5:/pt2:CL="c":

Figure 1 : CSD initialization



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PHILIPS IMS Interactive Media Systems	THE DOWNLOAD ROUTINE
CDI 605T System Software Rel 1.3 TECHNICAL MANUAL	
Version : 1.1 Chapter : 4.0 Status : Released	

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4.1 Downloading

4.2 Executing a downloaded object file

4.3 Downloading OS-9 modules

5 Definitions

6 Application notes

7 Example download 'session'

2 Introduction

The download routine offers the possibility to load data in the player memory from a host. The data loaded can be a test program which can be executed immediately or OS-9 memory modules that will be added to the module directory by the kernel later on during startup. Downloading is done via serial port#3 (console port).

Note : there will be no CRC check for downloaded modules !

3 Technical description

3.1 Position of download in boot

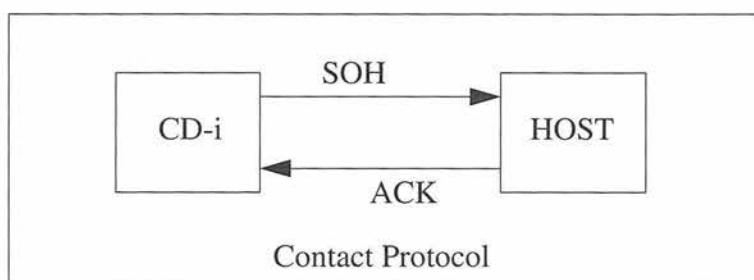
The download routine is activated early during startup before any messages have been printed and before the low level RAM/ROM search has been executed.

3.2 Communication with download routine

The communication with the download routine is made up of two parts, first making contact and after that sending commands and data.

3.2.1 Making contact

The download routine will send a SOH character (at 9600 baud) via the console port₍₁₎. If a correct response is received (ACK) within a specified period, the contact is established and the download routine will be able to accept commands and/or data. If the required response is not received, the normal startup will proceed. The time out period is very short (3 ms).



After the contact has been established, the communication speed is switched to 19200 baud. The HOST is now in control, the CD-i player will wait for commands and data sent by the HOST.

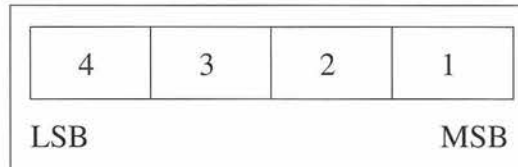
(1) : see also 'RS232 port specification', chapter 8.0

3.2.2 Sending commands

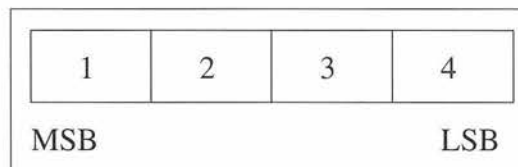
Commands are sent in packets. Each packet starts with a start byte (SOH) followed by a packet ID (one byte indicating the type of the packet). The data following these two bytes depends on the packet ID. The packet ends with a BCC character (parity byte) which is the XOR of all preceding bytes in the packet. The XOR of the whole packet (including the BCC character) should always be zero. Integer values (made up of more than one byte) are always sent with the MSB first (680x0 format).

Intel processors (in a PC) store the bytes just the other way around. A PC will therefore have to convert the values it sends to 680x0 format.

80x86: long



680x0: int



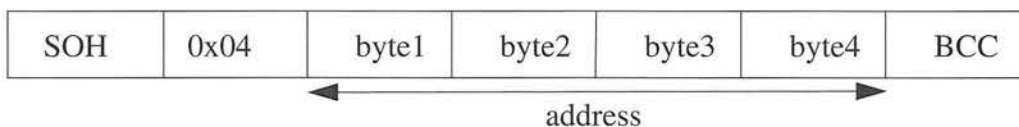
After a whole packet has been received correctly, the CD-i player will send an ACK character. When something has gone wrong, bad BCC character for example, the CD-i player will send a NAK character. If no character is sent at all after a packet has been received, you can be sure that something went wrong (a severe error may have occurred).

3.2.3 Possible commands

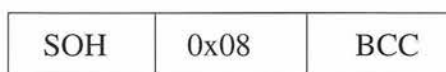
Set load address



Start execute



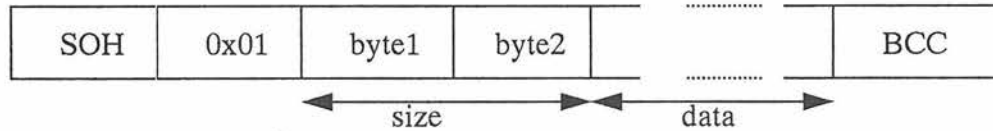
End download



Besides the above command packets, data packets may be sent.

The 'data' in a data packet is made up of the size of the data (two bytes) followed by the data itself.

Data packet



4 Possible usage

4.1 Downloading

To download data, a download address will have to be specified first. This is done with the "Set load address" command (see: Possible commands). After that, data packets can be sent. The data in these packets will be stored starting at the address specified in the "Set load address" command. If no download address is specified, the behaviour of the player is unpredictable.

4.2 Executing a downloaded object file

With the "Start execute" command, a downloaded program can be executed. This can be used to execute a raw object file. The download routine will treat the address specified in the "Start execute" command as the address of a subroutine and jump to it with a JSR instruction. If the program ends with an RTS instruction, control will return to the download routine. After the program has returned, the download routine will send an EM (end of message) character to indicate the program has ended.

4.3 Downloading OS-9 modules

When an "End download" command is received, the area that has been downloaded too will be added to the ROM list that is handed over to the kernel later on during startup. The kernel will then look for OS-9 memory modules in this area. It is also possible to download to different areas by sending a new "Set load address" command after some data has been downloaded. In that case, all areas that have been downloaded to will be added to the ROM list.

A consequence is that these areas will not be available any more as free RAM. The RAM lost will always be a multiple of the block size specified in the colored memory list in the init module (16Kb for video memory, 512Kb for extension memory).

After the "End download" command, the normal player startup procedure is continued. The communication speed is reset to the default value (9600 baud).

4.4 Starting an application

The download routine can be used to start an application as well. The application program module name must be "play" and it's revision number must be higher than the revision number of the default play module in ROM. When this application is loaded via the download routine, it will be started instead of the default application (player shell). This may be an alternative for loading small applications on CD-i players without a floppy drive.

5 Definitions

The character codes used for communication are all ASCII codes. They have been mentioned in the following table.

Name	Hex value	Meaning
SOH	\$01	start of heading
ACK	\$06	acknowledged
NAK	\$15	not acknowledged
EM	\$19	end of message

The host system is supposed to be a PC but it could be any other system as well. A program that communicates with the download routine in a CD-i player is available on a PC ("pccdi") and on a SUN ("cdiload").

6 Application notes

Not all memory in the player can be used for downloading. The areas 0x0 - 0x4000 and 0x80400 - 0x80500 should not be used, they are reserved for system software.

OS-9 memory modules could best be loaded starting at address 0x4000 (video plane A) or 0x80500 (video plane B). When extension memory is available, it can be used for downloading as well. The address of extension memory is fixed. See Chapter 11 for the Base Address.

7 Example download 'session'

For this example, a player with the following memory map is assumed:

0x000000 - 0x080000	video memory plane A
0x080000 - 0x100000	video memory plane B
0x600000 - 0x800000	extension memory

The following download packets are sent by the HOST:

Set load address

SOH	0x02	00	60	00	00	BCC
-----	------	----	----	----	----	-----

Player

ACK

Data packet

SOH	0x01	01	00	data	BCC
-----	------	----	----	------	-----

ACK

Data packet

SOH	0x01	01	00	data	BCC
-----	------	----	----	------	-----

ACK

Set load address

SOH	0x02	00	08	05	00	BCC
-----	------	----	----	----	----	-----

ACK

Data packet

SOH	0x01	00	80	data	BCC
-----	------	----	----	------	-----

ACK

End download

SOH	0x08	BCC
-----	------	-----

ACK

Each time a package is downloaded to the player, the player will answer to the host with an acknowledge.

The ROM list built by the download routine for the kernel will consist of the following areas:

address	size
0x600000	0x00200
0x080500	0x00080

Because these areas have been added to the ROM list, the available RAM will be reduced. The areas 0x600000-0x680000 (512 Kb) and 0x80000-0x80400 (16 Kb) will not be available anymore as free RAM after startup.

PHILIPS IMS Interactive Media Systems	INITIAL PROCESS SYSGO
CDI 605T System Software Rel 1.3 TECHNICAL MANUAL	
Version : 1.2 Chapter : 5.0 Status : Released	

1 Contents

2 Introduction

3 Starting up

4 Exiting a shell

2 Introduction

The last step in the boot procedure, is the startup of the first process. This process is defined by the user in the file SysGo.

3 Starting up

After the config program is forked to update the CSD₍₁₎, sysgo will wait for this process to terminate before it will open a path to the slave processor to get the start up mode. If an error occurred during forking the config program a reset will be generated. If an error occurred during opening the path or getting the startup mode, sysgo will continue its procedure as a normal start up mode. The possible startup modes are :

0 = normal mode

1 = service mode

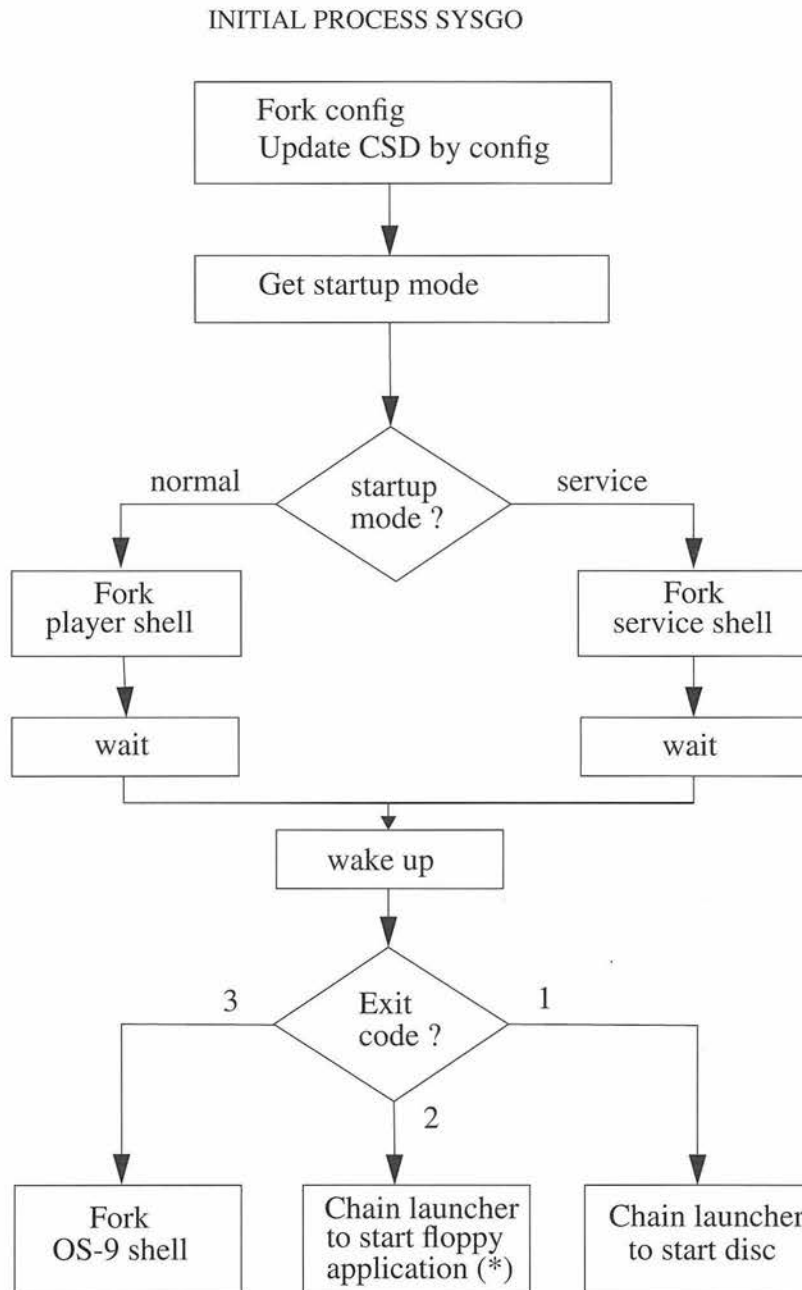
After the startup mode is known the player shell or the service shell is forked. Sysgo will go into a wait state and will now wait until a child process terminates. If an error occurs during forking one of the shells, an error message will be printed to the terminal (if connected to port#3) and the system will reset.

4 Exiting a shell

When an F\$Exit system call occurred an exit code is given. If the exit code equals 1 a CD-i application will be started by chaining the launcher. If the exit code equals 2 a floppy application will be started by chaining the launcher. If the code was 3 an OS-9 shell will be forked and sysgo will wait for the shell to terminate. If an error occurs during forking a shell or receiving the wrong exit code, a reset will be generated.

(1) see also chapter 3 : configuration status descriptor

Figure 1: Sysgo



Note : if an error occurs see description in text

(*) see chapter 6 : load application and system

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PHILIPS IMS Interactive Media Systems	LOAD APPLICATION and SYSTEM	
CDI 605T System Software Rel 1.3 TECHNICAL MANUAL		
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1 Contents

2. Introduction
3. Start OS-9 shell
4. Load floppy application

2 Introduction

In the settings menu of the player shell, two buttons exist : the 'load' button to start an application from floppy and the 'system' button to start an OS-9 shell or execute a batch file.

3 Start OS-9 shell.

When the player shell is in use and there is a terminal DSD '/term' in the CSD file₍₁₎, you can select "system" in the sub menu "settings".

- If you click once on "system", an OS-9 shell will start up and the player shell will be removed from memory.
- If you click twice on "system", the player shell tries to execute the batch file "/nvr/startup.prf". If the batch file is executed (or there was no such file) then the player shell continues in the settings screen.
- The player shell stays resident in memory during execution of the batch file, so beware of memory fragmentation.

The terminal must be initialized at 9600 baud, 8 bits/character, no parity.

4 Load floppy application.

When the player shell is in use, selecting the "load" option in sub menu "settings" will start an application from floppy. It works as follows: (see figure 1 and 2).

First of all, the player shell will test the kind of floppy that is inserted into the drive of the player by trying all the descriptors known by the shell (out of the CSD₍₁₎). If the drive does not correspond to one of these drivers the load instruction is cancelled after a certain time out₍₂₎.

Then the volume name is read from the floppy-disc and the two current directories (command and data) are set to the root directory of the floppy-disc in the drive. If this is not possible the load instruction is cancelled.

Next, the player shell searches the floppy-disc for an application file name the same as the volume label, and tries to load the application. If something goes wrong the load instruction is aborted.

(1) For more explanation see Configuration Status Descriptor chapter 3.0

(2) Note: Because of known problems, do not load a floppy application from a /d0uv, /pcd0 and /pcd0h floppy (See deviation list).

If the applications file name starts with "cdi_", the player shell tries to change the two current directories (command and data) to the CD-i disc's root directory by reading the entry in the CSD, so that the application program will behave as if it was loaded from the CD-i disc itself. If this change is not successful, the shell sets the command and data directory back to the root of the floppy-disc.

Then, the player shell retrieves the module name from the module header. Next, the player shell will start up the first module of the loaded file, thereby passing the parameters that followed the file name in the volume label, by chaining the player shell to this application. If something goes wrong during the startup of the application, the system will reset. If any error is returned (the load instruction is ignored), no further action is taken and the player shell waits for more input from the user.

E.g.

Volume name of the disc = 'discname'

The loadable file on disc has to have the same name 'discname'.

This file contains - Module1

- Module2

- Module3,...

When the load button is pressed in the settings menu, the shell will load the file with the same name as the volume name = 'discname' and will start Module1 (beware that the attributes of this file have to be set to executable!). Only Module1 is executed because this is the first module in the file 'discname' (data directory and execute directory point to the root of the floppy directory). If the applications file name starts with 'cdi_' both directories stand to the CD-i root's directory.

E.g. a tool :

Exebat is a program that enables starting a batch file from a /d0 diskette with the load option. In exebat the command : 'system ("/d0/startup")' serves to start up the batch file 'startup'.

Conditions :

- volume name has to be 'exebat' (only available for one floppy format : /d0).
- the file 'exebat' must be present in the root directory.
- the name of the batchfile must be 'startup' and must be in the root directory too.

Startup is the batch file and may consist of the following :

- built in shell commands.
- names of other existing batch files.
- names of existing executable modules.

e.g. Startup can consist of the following :

```
chx /d0/cmds
load -d module1 module2 module3
module1
module2
unlink module1
module3
unlink module2
unlink module3
```

Note : there has to be a shell available in memory.

All commands in the batchfile are executed consecutively one by one.

Figure 1:

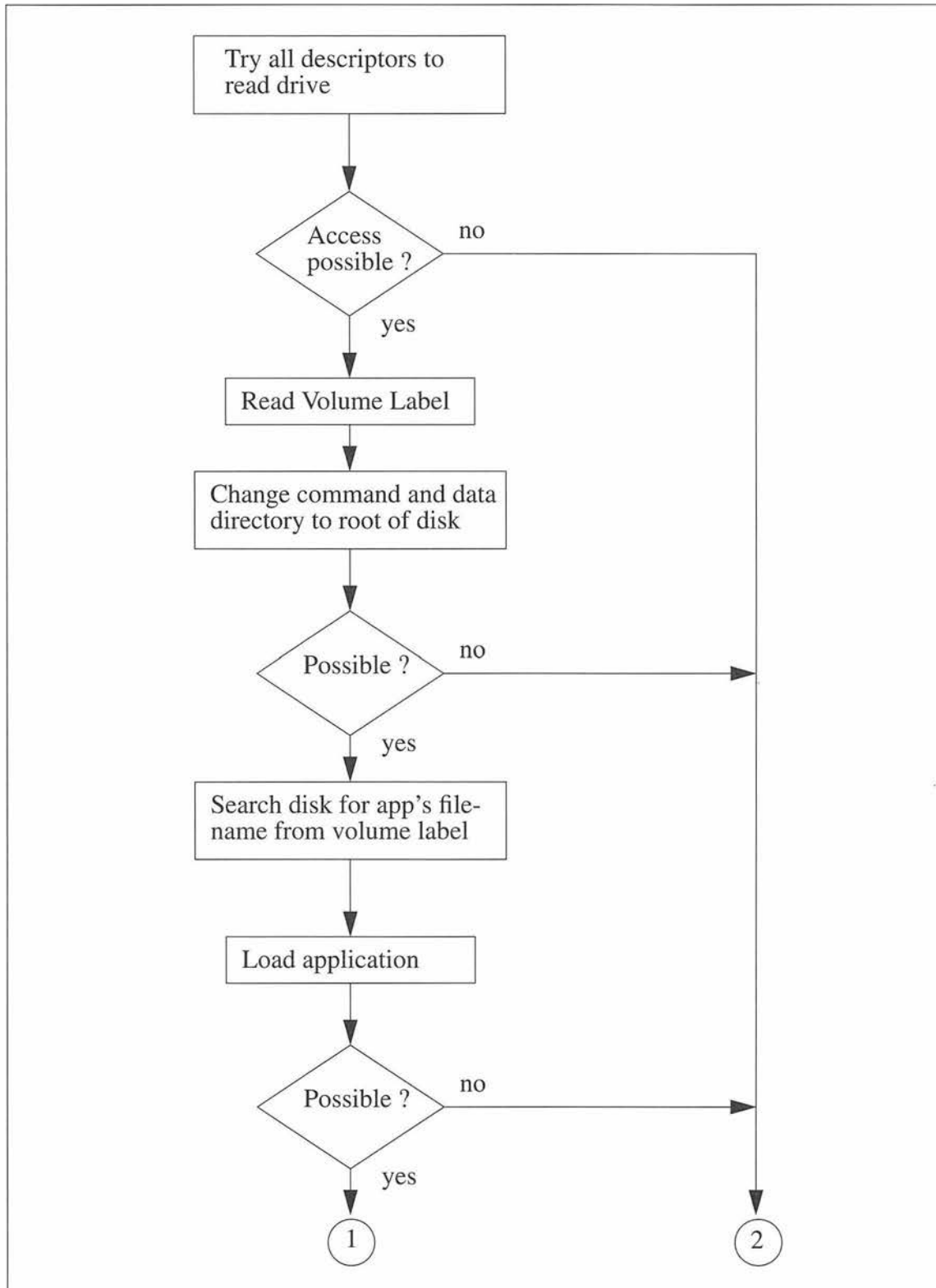
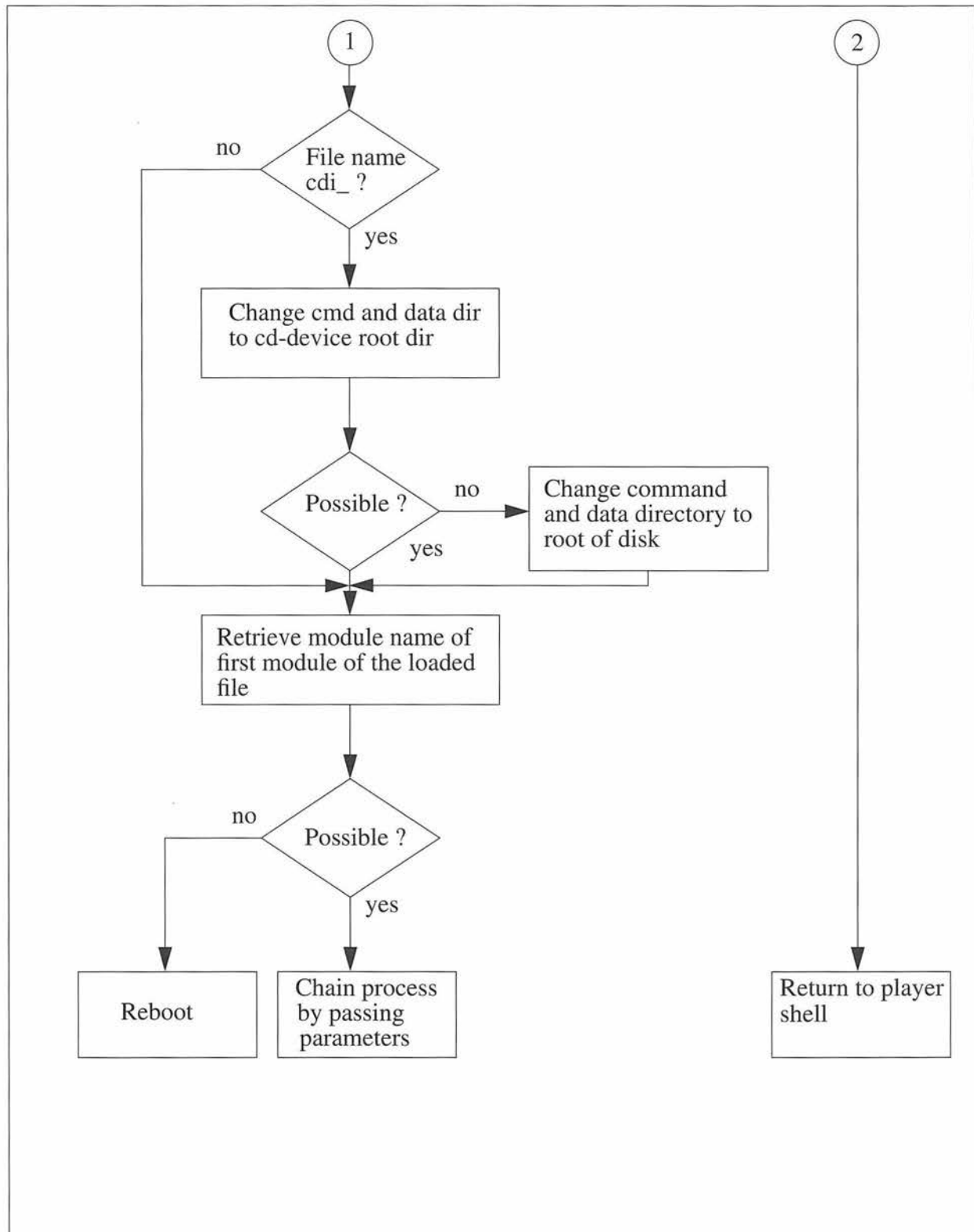


Figure 2:



PHILIPS IMS Interactive Media Systems	TECHNICAL CHARACTERISTICS OF THE PLAYER
CDI 605T System Software Rel 1.3 TECHNICAL MANUAL	
Version : 1.1 Chapter : 7.0 Status : Released	

1 Contents

2. General

3. Extension

3.1 I/O functions

3.2 Memory

2 General

AUDIO:	L+R (Cinch connector) 2 V rms typical 10 KOhm. Headphone (6 mm Headphone socket). 8 Ohm - 2 KOhm, 30 mW at 32 Ohm. Manual volume control.
VIDEO:	NTSC and PAL video out: - RGB (D-sub 9 connector) 0.7 Vp-p into 75 Ohm. (0.3 Vp-p into 75 Ohm sync). - Y/C (Y/C connector) 1 Vp-p into 75 Ohm. - CVBS (Cinch connector) 1 Vp-p into 75 Ohm.
CONTROL PORTS:	Port 1 (mini DIN 8-pin connector) for pointing device, or CD-i keyboard. Port 2 (mini DIN 8-pin connector) for pointing device, CD-i keyboard or RS232-C based peripherals. Port 3 (D-sub 9 connector) for RS232-C based peripherals. Terminal/Console port. Port 4 (D-sub 9 connector) for RS232-C based peripherals.
EXTENSION SLOT:	One general slot for system extensions such as SCSI (hard disc), Ethernet, Centronics port, additional memory (4 Mbyte). The second extension slot is prepared to add the Full-Motion Video function.
REMOTE CONTROL: (OPTIONAL)	- Infra-red remote control handset. - Wired infra-red remote control receiver. - RC5 extended as well as RC6 compatible.
EMULATOR:	- Digital out in. - RS out. Used as emulation for CD-I application development.
MEMORY:	- 2 x 512 Kb Base Case RAM + 4 Mb RAM. - 8 Kb non volatile RAM. - 3.5 inch floppy disc drive. - 512 Kb ROM for Base Case. - 512 Kb ROM for Extension.
	<u>Standard CD-i systems normally have slidely more RAM then the 2X480k. To avoid problems during application/title development it is adviced to use only colored memory.</u>
CPU:	68070 at 15 Mhz.

3 Extension

The CDI 605T extension can be seen as a multi function I/O and memory board. On this board the following functions are available:

- I/O: - Parallel printer port.
- SCSI port.
- Ethernet port.
- Memory : - EPROM/ROM.
- DRAM.

3.1 I/O functions

- *Parallel printer port :*

- connector: 25 pins subd connector.
- standard: centronics compatible.
- function: serves as a printer interface. It has the same functionality as the parallel printer interface used in a pc-at. With dedicated software it can be used as a general purpose i/o port.

- *SCSI port :*

- connector: 50 pins microribbon connector.
- standard: SCSI-1 (8 bit).
- function: serves as an interface for all SCSI based devices, such as harddisks, CDROMs, magnetic tapes, ...
Depending on the (resident) software installed, the system (player + ext) can act as a master (i.e. HARDDISK) as well as a slave.

- *Ethernet :*

- connector : BNC connector (thin wire).
- standard : IEEE 802.3 (thin ethernet).
- function : serves as an interface to all thin wire based LAN's.

3.2 Memory

- *RAM :*

- function : it serves as a memory extension for the CD-i player. It provides an extra 4 Mbyte of additional system memory.

- note: Once the extension is plugged in, the CD-i system will automatically add the extra RAM to its already available system RAM.

- *ROM/EPROM* :

function : It provides the CD-i player with the necessary software for all previous functions (Parallel printer, SCSI, Ethernet).

Software available :

- printer : Driver + Descriptor (SCF device).
- SCSI : Harddisk SCSI driver + Descriptor (RBF device) + Tape streamer.
- Ethernet: TCP/IP software package (telnet, ftp).
NFS software package (optional).
- memory : none.

PHILIPS IMS Interactive Media Systems	SERIAL PORT SPECIFICATION
CDI 605T System Software Rel 1.3 TECHNICAL MANUAL	
Version : 1.1 Chapter : 8.0 Status : Released	

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2. Scope
3. Technical specification
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 - 3.2 Port #2
 - 3.2.1 Signals & pinning for RS232-format
 - 3.3 Port #3 & port #4
4. Functionality and port behavior
5. Appendix A: port device in relation with the CSD

2 Scope

This document gives the specification of the ports on a CD-i player. Each port is described in detail below. This document only gives the hardware specification, the software specification is given in the document : 'Input devices' ⁽¹⁾ and SCF documentation.

Port1 and Port2 can accept RS232-C as well as TTL signals.

Port3 and Port4 are always RS232-C level based.

(1) Input devices : chapter 9 of the technical manual

3 Technical specification.

3.1 PORT #1

- Connector type : 8 pins miniDIN (female).
- Location : front.
- Usage : for standard CD-i pointing devices and keyboard (RS232 or TTL).
- Internal device: Slave proc.(68HC05, SCI).
- Buffers: transmitter: 1 byte receiver: 1 byte.
transmit & receive shift register not included.
(transmitter cannot be used).
- Baudrates : 1200 and 9600.
- Dataformat : for pointing devices : 1 startbit, 7 databits, 2 stopbits, no parity.
for keyboards : 1 startbit, 8 databits, 1 stopbit, no parity.
- Handshake : hardware : only RTS (see 'Input devices' ⁽¹⁾).

(1) Input devices : chapter 9 of the technical manual

3.1.1 Signals & Pinning for RS232-format:

<u>pin</u>	<u>Signal</u>	<u>Description</u>	<u>Inp/Outp.</u>
1	n.c.	Not Connected	
2	RXD	Received Data	Input (**)
3	TXD	Transmitted Data (not used)	Output
4	DTR (*)	Data Terminal Ready	Output
5	GND	Signal Ground	
6	n.c.	Not Connected	

7	RTS	Request To Send	Output
8	+5V	Power supply for the device	

Input : this is an input signal for the CD-i player.

Output: this is an output signal for the CD-i player.

(*) : DTR can not be used as an active handshake signal. It serves as a power supply for some input devices.

(**): These inputs are TTL level compatible.

All signal levels and current ratings, except +5V and DTR are according EIA RS232-C standard.

RATINGS: +5V: +5V +/-10% max. 200mA.

DTR: +12V +/-10% (no load) max. 10mA (Ri = 1k).

TTL level definitions are shown below (only for RXD):

<u>Signal level (TTL)</u>	<u>$\leq +0.8\text{ V}$</u>	<u>$\geq +2\text{ V}$</u>
Binary state (for DATA)	1	0
Signal Condition	Mark	Space
Control & Timing Function	OFF (Negated)	ON (Asserted).

3.2 PORT #2

- Connector type : 8 pins miniDIN (female).
- Location : rear.
- Usage : for standard CD-i pointing devices, keyboard and other RS232 or TTL devices (see 'Input devices' (1)).
- Internal device : on chip UART (SCC68070).
- Buffers : transmitter: 1 byte receiver : 1 byte.
transmit & receive shift register not included.
- Baudrates : 75, 150, 300, 1200, 2400, 4800, 9600 and 19200.
Independently programmable baudrate for receiver and transmitter.
- Dataformats : character length : 7 or 8 bits.
stopbit length : 1 or 2 bits.
parity : odd, even or no.
- Handshake : hardware : 2 control lines ; RTS and CTS.
software : XON/XOFF.

(1) Input devices : see chapter 9 of the technical manual

3.2.1 Signals & Pinning for RS232-format:

<u>pin</u>	<u>Signal</u>	<u>Description</u>	<u>Inp/Outp</u>
1	n.c.	Not Connected	
2	RXD	Received Data	Input (**)
3	TXD	Transmitted Data	Output
4	DTR (*)	Data Terminal Ready	Output
5	GND	Signal Ground	
6	CTS	Clear To Send	Input (**)
7	RTS	Request To Send	Output
8	+5V	Power supply for the device	

Input : this is an input signal for the CD-i player.

Output: this is an output signal for the CD-i player.

(*) : DTR is not an active handshake signal. It serves as a power supply for some input devices.

(**): These inputs are TTL level compatible.

All signal levels and current ratings, except +5V and DTR, are according the EIA RS232-C standard.

RATINGS: +5V: +5V +/-10% max. 200mA.

DTR: +12V +/-10% (no load) max. 10mA (Ri = 1k).

TTL level definitions are shown below (only for RXD):

<u>Signal level (TTL)</u>	<u>< +0.8V</u>	<u>>+2V</u>
Binary state (for DATA)	1	0
Signal Condition	Mark	Space
Control & Timing Function	OFF (Negated)	ON (Asserted)

3.3 PORT #3 & PORT #4

- Connector type : 9 pins sub-D (male).
- Location : rear.
- Usage : for RS232 devices.
- Internal device:
 - PORT #3 = DUART (68681) port A.
 - PORT #4 = DUART (68681) port B.
- Buffers : transmitter : 1 byte receiver : 3 bytes.
transmit & receive shift register not included.
- Baudrates : 50, 75, 110, 134.5, 150, 300, 600, 1200, 1800, 2400 4800, 7200, 9600, 19200 and 38400.
Independently programmable baudrate for receiver and transmitter.
- Dataformats : character length : 5, 6, 7 or 8 bits.
stopbit length : 0.5 to 2 bits.
parity : odd, even or no.
- Handshake :
 - hardware : 2 control lines; RTS and CTS.
 - software : XON/XOFF.

Signals & Pinning:

<u>pin</u>	<u>Signal</u>	<u>Description</u>	<u>Inp/Outp.</u>
1	n.c.	Not Connected	
2	RXD	Received Data	Input
3	TXD	Transmitted Data	Output
4	n.c.	Not Connected	
5	GND	Signal Ground	
6	n.c.	Not Connected	
7	RTS	Request To Send	Output
8	CTS	Clear To Send	Input
9	n.c.	Not Connected	

Input : this is an input signal for the CD-i player.

Output: this is an output signal for the CD-i player.

4 **Functionality and port behavior** ⁽¹⁾

- If a pointing device is connected only to port#1, it will become the first (or default) CD-i pointing device.
- If a pointing device is connected to port#1 and one to port#2, both devices will become available as pointing devices for the application. The device connected to port#1 will become the first pointing device, the one connected to port#2 will become the second pointing device.
- If a pointing device is connected only to port#2, it will become the first CD-i pointing device. In this case the remote control as well as a pointing device connected to port#1 during operation, will act as the second pointing device (see notes).
- If no pointing devices are connected the remote control is considered as the first pointing device. In this situation a pointing device can be added during operation to port#1 only. It will become also the first pointing device (see notes).
- If a keyboard is connected only to port#1, it will become the first CD-i keyboard.
- If a keyboard is connected only to port#2, it will become the first CD-i keyboard.
- If a keyboard is connected to port#1 and one to port#2, both keyboards will become available as keyboards for the application. The keyboard connected to port#1 will become the first keyboard, the one connected to port#2 will become the second keyboard.
- If port#2 does not contain a keyboard, modem, nor a pointing device it will be considered as an SCF device.
- During operation a pointing device connected to port #1 may always be pulled out, plugged in or replaced by an other pointing device (not by a keyboard !).
- Port#3 and port#4 will always be general RS232 interfaces.

Notes:

- remote control: The remote control is always available as a pointing device for the application. Port#1 and the remote control have the same path to the application.
- first/second pointing device: First pointing device means : consecutively the first pointing device DSD in the CSD file. Second pointing device means : consecutively the second pointing device DSD in the CSD file. The same is valid for first/second keyboard.
- port#2: If one wants to use a device (pointing, keyboard or modem) on port#2, the device should always be connected before power on or before a system restart.
- Keyboard: Till edition 7 of the module config, only port#2 was supporting a keyboard. With editions 7 and higher, also port#1 can support a keyboard.

(1) See also chapter 03 : Configuration Status Descriptor

5 Appendix A : Port device in relation with the CSD

Each of the devices connected to one of the available ports are related to a specific descriptor. Below we see the relationship between a device, the port it is connected to (at power-on) and it's DSD in the CSD.

1) PORT#1 (handled by the slave) :

Uni-directional

No device (only remote control)	: 5:/ptr:CL="c":
Mouse, Trackerball	: 5:/ptr:CL="c":
Joystick	: 5:/ptr:CL="d":
Graphic tablet	: 5:/ptr:CL="b":
Touch screen	: 5:/ptr:CL="a":
Keyboard	: 10:/kb1:KG="ALL":LA#0:RP#0:
Extended keyboard	: 10:/kb1:KG="1,2,3,4,5,7":LA#0:RP#0:

2) PORT#2 (68070 uart) :

Uni-directional

No device	: -
Mouse, Trackerball	: 5:/pt2:CL="c":
Joystick	: 5:/pt2:CL="d":
Graphic tablet	: 5:/gt12:CL="b": or 5:/gt96:CL="b": (1)
Touch screen	: 5:/gt12:CL="a": or 5:/gt96:CL="a": (1)
Keyboard	: 10:/kb:KG="ALL":LA#0:RP#0:
Extended keyboard	: 10:/kb:KG="1,2,3,4,5,7":LA#0:RP#0:

Bi-directional

SCF device	: 20:/t2:
Modem device	: 20:/t2:CL="MDM":

(1) : The descriptor gt12 is used for 1200 baud, the descriptor gt96 for 9600 baud.

3) PORT# 3 (68681 duart port A) :

Bi-directional

SCF device (terminal) : 20:/term:

4) PORT# 4 (68681 duart port B):

Bi-directional

SCF device : 20:/t1:

Notes:

- The DSD's described above are placed in the CSD by editions #14 and higher of the module config. **With editions #14 and higher of config the CSD is now at power-on completely conform the green book.** This was not the case with editions #13 and lower of config:
 - No supporting of extended keyboards.
 - PORT#1 : the DSD was 5:/ptr:CL="c"; for all pointing devices.
PORT#2 : the DSD for a joystick was the same as for a mouse (5:/pt2:CL="c";), the DSD for a touch screen was the same as for a graphic tablet (5:/gt12:CL="b"; or 5:/gt96:CL="b";).
- Because of the possibility to replace a pointing device connected to port#1 during operation and the fact that the CSD is only updated at power-on, it is understandable that for port#1 the CSD can deviate from the green book during operation. For port#2 the CSD stays conform the green book.

Note : The modem entry is set in the CSD by the config module with edition #15 and higher.

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CDI 605T System Software Rel 1.3 TECHNICAL MANUAL		
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2 Introduction

This document describes some pointing devices, keyboards and modems for a CD-i player. For each device the most important items will be discussed :

- The identification.
- The data format.
- The data flow.

Appendix B gives the relation between device and CSD.

The input device should be equipped with the suitable male connector.

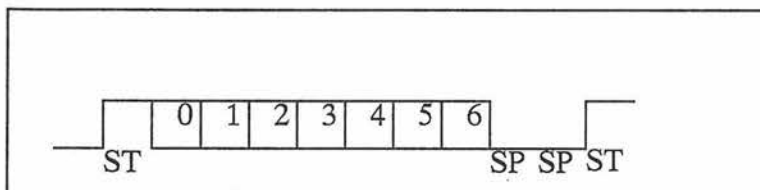
3 Data Signals

For the communication the RS232C or TTL format will be used.

For signal name description and standard see chapter 8.0 : 'Serial Port Specification' (see also EIA RS232-C and TTL standard).

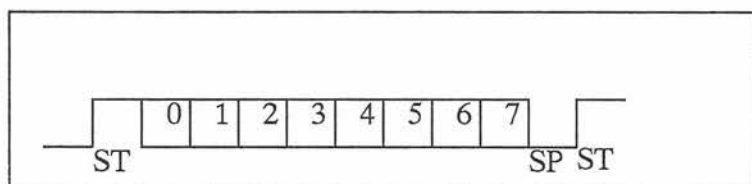
3.1 Data signals for pointing devices

- Baudrate : 1200 or 9600 baud.
- Data format :
 - 1 start bit (ST).
 - 7 data bits (LSB first).
 - 2 stop bits (SP).



3.2 Data signals for keyboards

- RS232C or TTL format: 1200 baud.
- Data format:
 - 1 start bit (ST).
 - 8 data bits (LSB first).
 - 1 stop bit (SP).
 - no parity.



Signals : During normal operation the RTS signal should be asserted (logic 0 or ON). The data from input device to CD-i player should always be sent via pin 2 (RXD).

Buffer : 16 bytes (in the keyboard).

3.3 Data signals for modems (only on port 2)

Mini Din 8a type.

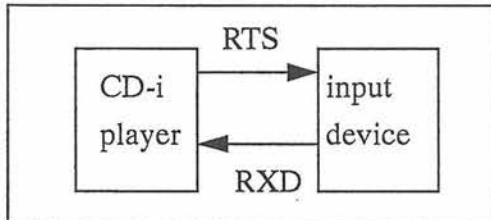
<u>pin nr.</u>	<u>Signal</u>	<u>Description</u>	<u>Inp/Outp.</u>
1	--		
2	RXD	Received Data	Input
3	TXD	Transmitted Data	Output
4	--		
5	GND	Signal Ground	
6	CTS	Clear to send	Input
7	RTS	Request to send	Output
8	+ 5 Volt	Power supply for device	

Note: -- means not used by TeleCD-adapter.

See also TeleCD ToolKit

3.4 Name convention

For identification of pointing devices, keyboards and modems there are only two signals which will carry data.



The first is the RTS signal and is always an output for the CD-i player. The second is the RXD signal and is always an input for the CD-i player.

- RXD : Carries the data from pointing device, keyboard or modem to player.
- RTS : Will be used for device identification (see 4. device recognition).

The other signals can be used by the pointing device or keyboard to get power supply from the CD-i player (see appendix A).

4 Device Recognition

To be able to identify the input device that is connected to the CD-i player, every input device will transmit its own identification (ID) byte on request.

To maintain compatibility over various products the device recognition can occur at two speeds: 1200 and 9600 Bps, selection of the speed must be possible, if not selectable then the speed of the device must be set to 1200 Bps.

- 1 start bit
- 7 data bits (LSB first)
- 1 stop bit.

4.1 Device ID sequence

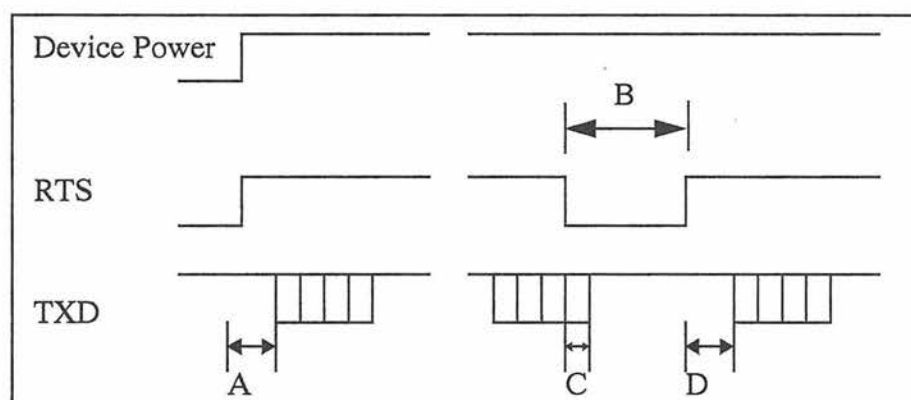
1. Negate RTS: Current data transmission to the CD-i player is stopped.
2. Assert RTS.
3. The device will send its 1 byte device ID.

4. It is recommended to send a complete data packet after the identification byte to know the state of the buttons at start-up.

4.2 Device ID codes

	<u>d6</u>	<u>d5</u>	<u>d4</u>	<u>d3</u>	<u>d2</u>	<u>d1</u>	<u>d0</u>	
Mouse/Trackerball :	1	0	0	1	1	0	1	= ascii char. 'M' (0x4D)
Joystick :	1	0	0	1	0	1	0	= ascii char. 'J' (0x4A)
Graph.tab.:	1	0	1	0	1	0	0	= ascii char. 'T' (0x54)
Touch scr.:	1	0	1	0	0	1	1	= ascii char. 'S' (0x53)
Modem :	0	1	1	0	0	0	0	= ascii char. 'O' (0x30)

4.3 Device ID timing



A : device power-on to ID byte timing:

Min. 100ms
Max. 500ms.

B : negate RTS time:

Min. 10ms
Max. irrelevant.

C : Continued data transmission time when RTS is negated:

Min. 0ms
Max. 1 byte time (1200 baud = 8.3ms, 9600 baud = 1ms).

D : assert RTS to ID byte timing when device is powered:

Min. 1 bit time (1200 baud = 866 μ s, 9600 baud = 104 μ s).
Max. 10 ms.

5 MOUSE/TRACKERBALL specification

5.1 Data flow

During normal operation (RTS asserted) :

- Data is sent to the CD-i player via RXD.
- Every time there is a mouse movement or a button state change the three byte mouse data is transmitted from mouse to CD-i player.
- When the mouse is not moved and the buttons do not change state (they stay as they are, pressed or released), it should not send any 3 byte mouse data to the CD-i player.
- An on going data packet transmission should never be interrupted by a button state change or a mouse position change.

5.2 Data Format

Baudrate : 1200 baud.

	<u>d6</u>	<u>d5</u>	<u>d4</u>	<u>d3</u>	<u>d2</u>	<u>d1</u>	<u>d0</u>
Byte 0	1	bt1	bt2	Y7	Y6	X7	X6
Byte 1	0	X5	X4	X3	X2	X1	X0
Byte 2	0	Y5	Y4	Y3	Y2	Y1	Y0

- bt1 = button1 : 1 = pressed 0 = not pressed.
- bt2 = button2 : 1 = pressed 0 = not pressed.
- X7..X0 :
 - 8 bit data horizontal movement.
 - 2's complement.
 - left direction = negative (\$FF..\$80).
 - right direction = positive (\$01..\$7F).
- Y7..Y0 :
 - 8 bit data vertical movement.
 - 2's complement.
 - up direction = negative (\$FF..\$80).
 - down direction = positive (\$01..\$7F).

A databyte = \$00 indicates that there was no movement in that direction.

6 JOYSTICK specification

6.1 Data flow

During normal operation (RTS asserted) :

- Data is sent to the CD-i player via RXD.
- Only when the stick is moved out of its center position, a 3 byte data packet is sent continuously to CD-i player.
- When the stick is released (center position) and the buttons do not change state (they stay as they are, pressed or released), the joystick should not send any 3 byte joystick data to the CD-i player.
- Every time a button changes state, a 3 byte data packet is sent to the CD-i player.
- An on going data packet transmission should never be interrupted by a button change or a stick position change.

6.2 Data Format

Baudrate : 1200 baud.

	<u>d6</u>	<u>d5</u>	<u>d4</u>	<u>d3</u>	<u>d2</u>	<u>d1</u>	<u>d0</u>
Byte 0	1	bt1	bt2	Y7	Y6	X7	X6
Byte 1	0	X5	X4	X3	X2	X1	X0
Byte 2	0	Y5	Y4	Y3	Y2	Y1	Y0

- bt1 = button1 : 1 = pressed 0 = not pressed.
- bt2 = button2 : 1 = pressed 0 = not pressed.
- X7..X0 :
 - 8 bit data horizontal movement.
 - 2's complement.
 - left direction = negative (\$FF..\$80).
 - right direction= positive (\$01..\$7F).
- Y7..Y0 :
 - 8 bit data vertical movement.
 - 2's complement.
 - up direction = negative (\$FF..\$80).
 - down direction = positive (\$01..\$7F).

A databyte = \$00 indicates that there was no movement in that direction.

See Appendix A : remarks about Joysticks for a CD-i player.

- Y9..Y0 :
 - 10 bit absolute vertical position.
 - outermost upper position: \$000.
 - outermost lower position: \$3FF.

Note : for touch screen bt1, bt2 = same and pen down is not relevant.

8 KEYBOARD specification

8.1 Keyboard ID specification

To be able to identify that a keyboard is connected to the CD-i player, it will send its own ID on request.

- Keyboard ID sequence:
 - Negate RTS: Current data transmission to the CD-i player is stopped.
 - Assert RTS.
 - The keyboard will send its 1 byte device ID, and clear its buffer.
 - Data transmission to the CD-i player can continue. The first byte sent to the CD-i player should always be the header byte of the device data packet. (The header byte is the first byte of a data packet d7 = 1).
- Keyboard ID code:

	<u>d7</u>	<u>d6</u>	<u>d5</u>	<u>d4</u>	<u>d3</u>	<u>d2</u>	<u>d1</u>	<u>d0</u>	
Keyboard :	1	1	0	0	1	0	1	1	= ascii char 'K'
Extended keyboard :	1	1	0	1	1	0	0	0	= ascii char 'X'

8.2 Keyboard Data Format

- Data flow during normal operation:
 - Every time there is a keyboard buttons state change the keyboard will send its current state (last pressed key code or release key code and the special key (*) status) in a 2 byte data packet to the CD-i player via pin 2 of the connector.

(*) special keys : Shift, Capslock, Supershift and Control.

- If there is no button state change, the keyboard should not send any 2 byte keyboard data to the CD-i player.
- If one or more of the special keys only are pressed or released, the keyboard will send the 2 byte data packet with its current special key status, extension bits = 1 and Key-code = \$00.
- Every time a key is released (no buttons pressed), the keyboard will send following data:

S3-S0 : 0000

M1-M0 : 01

K7-K0 : 00000000

This gives the following result : first byte \$82 second \$00.

- Data Format:

	<u>d7</u>	<u>d6</u>	<u>d5</u>	<u>d4</u>	<u>d3</u>	<u>d2</u>	<u>d1</u>	<u>d0</u>
1st	1	S3	S2	S1	S0	M1	M0	K7
2nd	0	K6	K5	K4	K3	K2	K1	K0

- S0-S3 : Status of the special keys :
 - S0 : Shift button(s) pressed when set.
 - S1 : CapsLock = on when set.
 - S2 : Supershift button(s) pressed when set.
 - S3 : Control button(s) pressed when set.
- M0-M1 : Extension bits :
 - 0 0 : Standard Character set (Green Book).
 - 0 1 : Character set 1 (if the keycode = \$00 then only special keys are pressed).
 - 1 0 : Character set 2 (Future use).
 - 1 1 : Character set 3 (Future use).
- K0-K7 Key-code. According to character- set ISO 8859-1 or an extended character set depending on the M0 and M1 bits.
- Data Range (M&K bits):
 - \$000 - \$0FF : Standard character set (FFGB VI-3).
 - \$100 : Keycode if key released Keycode for special key(s) status.
 - \$101 - \$1FF : Character set 1.
 - \$200 - \$2FF : Character set 2 (Future use).
 - \$300 - \$3FF : Character set 3 (Future use).
- Function CAPSLOCK :
 - Capslock only affects the alphabetical keys.
 - Capslock = off : normal = lower case (\$61-\$7A) shifted= upper case (\$41-\$5A).
 - Capslock = on : normal = upper case (\$41-\$5A) shifted= lower case (\$61-\$7A).

8.3.2 Keyboard character code table USA English version

Keyboard state : CAPSLOCK = off, so Normal = lower case

NR	Normal char	Normal word	Shift char	Shift word	SShift word	Ctrl word	SS+Ctrl word	S+SS word
1	[]	60	[~]	7E	E0	60	E0	FE
2	[1]	31	[!]	21	B1	31	B1	A1
3	[2]	32	[@]	40	B2	00	80	C0
4	[3]	33	[#]	23	B3	33	B3	A3
5	[4]	34	[\$]	24	B4	34	B4	A4
6	[5]	35	[%]	25	B5	35	B5	A5
7	[6]	36	[^]	5E	B6	1E	9E	DE
8	[7]	37	[&]	26	B7	37	B7	A6
9	[8]	38	[*]	2A	B8	38	B8	AA
10	[9]	39	[()]	28	B9	39	B9	A8
11	[0]	30	[)]	29	B0	30	B0	A9
12	[-]	2D	[_]	5F	AD	1F	9F	DF
13	[=]	3D	[+]	2B	BD	3D	BD	AB
14	[`]	5C	[~]	7C	DC	5C	DC	FC
15	delete	7F		0F	0F	1F	0F	0F
16	tab	09		19	19	19	19	19
17	[q]	71	[Q]	51	F1	11	91	D1
18	[w]	77	[W]	57	F7	17	97	D7
19	[e]	65	[E]	45	E5	05	85	C5
20	[r]	72	[R]	52	F2	12	92	D2
21	[t]	74	[T]	54	F4	14	94	D4
22	[y]	79	[Y]	59	F9	19	99	D9
23	[u]	75	[U]	55	F5	15	95	D5
24	[i]	69	[I]	49	E9	09	89	C9
25	[o]	6F	[O]	4F	EF	0F	8F	CF
26	[p]	70	[P]	50	F0	10	90	D0
27	[l]	5B	[L]	7B	DB	5B	DB	FB
28	[;]	5D	[:]	7D	DD	5D	DD	FD
30	ctrl	00		00	00	00	00	00
31	[a]	61	[A]	41	E1	01	81	C1
32	[s]	73	[S]	53	F3	13	93	D3
33	[d]	64	[D]	44	E4	04	84	C4
34	[f]	66	[F]	46	E6	06	86	C6
35	[g]	67	[G]	47	E7	07	87	C7
36	[h]	68	[H]	48	E8	08	88	C8
37	[j]	6A	[J]	4A	EA	0A	8A	CA
38	[k]	6B	[K]	4B	EB	0B	8B	CB
39	[l]	6C	[L]	4C	EC	0C	8C	CC
40	[.]	3B	[,]	3A	BB	3B	BB	BA

KEYBOARD specification

41	[']	27	["]	22	A7	27	A7	A2
43	return	0D		0D	0D	0D	0D	0D
44	shift	00		00	00	00	00	00
46	[z]	7A	[Z]	5A	FA	1A	9A	DA
47	[x]	78	[X]	58	F8	18	98	D8
48	[c]	63	[C]	43	E3	03	83	C3
49	[v]	76	[V]	56	F6	16	96	D6
50	[b]	62	[B]	42	E2	02	82	C2
51	[n]	6E	[N]	4E	EE	0E	8E	CE
52	[m]	6D	[M]	4D	ED	0D	8D	CD
53	[,]	2C	[<]	3C	AC	2C	AC	BC
54	[.]	2E	[>]	3E	AE	2E	AE	BE
55	[/]	2F	[?]	3F	AF	2F	AF	BF
57	shift	00		00	00	00	00	00
58	supersh.	00		00	00	00	00	00
61	space	20		20	A0	20	A0	A0
64	capslock	00		00	00	00	00	00
65	F2	81		89	91	99	91	91
66	F4	83		8B	93	9B	93	93
67	F6	85		8D	95	9D	95	95
68	F8	87		8F	97	9F	97	97
69	help	1C		1C	1C	1C	1C	1C
70	F1	80		88	90	98	90	90
71	F3	82		8A	92	9A	92	92
72	F5	84		8C	94	9C	94	94
73	F7	86		8E	96	9E	96	96
74	menu	1D		1D	1D	1D	1D	1D
90	esc	1B		1B	1B	1B	1B	1b
91	[7]	37		37	B7	37	B7	B7
92	[4]	34		34	B4	34	B4	B4
93	[1]	31		31	B1	31	B1	B1
95	home	1E		1E	1E	1E	1E	1E
96	[8]	38		38	B8	38	B8	B8
97	[5]	35		35	B5	35	B5	B5
98	[2]	32		32	B2	32	B2	B2
99	[0]	30		30	B0	30	B0	B0
100	culeft	08		11	15	02	15	15
101	[9]	39		39	B9	39	B9	B9
102	[6]	36		36	B6	36	B6	B6
103	[3]	33		33	B3	33	B3	B3
104	[.]	2E		2E	AE	2E	AE	AE
105	curoleft	0C		14	18	05	18	18
106	cucup	0B		13	17	04	17	17
107	cudown	0A		12	16	03	16	16
108	return	0D		0D	0D	0D	0D	0D

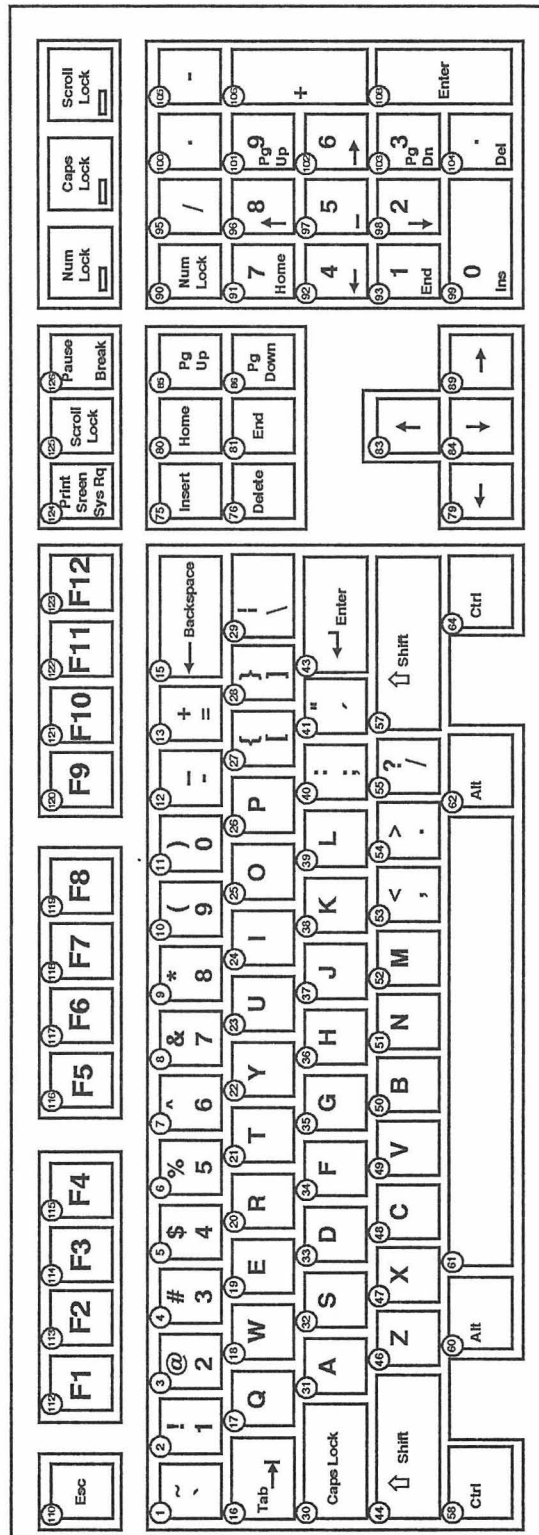
8.4 Implementation of an Extended keyboard

The behavior of the extended keyboard is the same as the USA English version keyboard except for following items :

- NUMLOCK function; it only affects the keys : 91..93, 96..99 and 101..104.
Numlock=off: as described in 8.4.2.
Numlock=on: as described in 8.4.3.
- More Keys are available:
Print Screen, SysReq, ScrollLock, Pause, Break, PC-Delete, Insert, PageUp, PageDown, End, F9, F10, F11 and F12.
- The keyboard can use its own (special) keys to generate some other character codes. (these keys are a local matter for the keyboard).

Examples: Numlock, AltGraphics,...

8.4.1 Keyboard Layout:



8.4.2 Keyboard character code table 1

Keyboard state: CAPSLOCK = off, so Normal = Lower case, NUMLOCK = off

NR	Normal char	Normal word	Shift char	Shift word	SShift word	Ctrl word	SS+Ctrl word	S+SS word
1	[~]	60	[~]	7E	E0	60	E0	FE
2	[1]	31	[!]	21	B1	31	B1	A1
3	[2]	32	[@]	40	B2	00	80	C0
4	[3]	33	[#]	23	B3	33	B3	A3
5	[4]	34	[\$]	24	B4	34	B4	A4
6	[5]	35	[%]	25	B5	35	B5	A5
7	[6]	36	[^]	5E	B6	1E	9E	DE
8	[7]	37	[&]	26	B7	37	B7	A6
9	[8]	38	[*]	2A	B8	38	B8	AA
10	[9]	39	[()]	28	B9	39	B9	A8
11	[0]	30	[)]	29	B0	30	B0	A9
12	[-]	2D	[_]	5F	AD	1F	9F	DF
13	[=]	3D	[+]	2B	BD	3D	BD	AB
15	delete	7F		0F	0F	1F	0F	0F
16	tab	09		19	19	19	19	19
17	[q]	71	[Q]	51	F1	11	91	D1
18	[w]	77	[W]	57	F7	17	97	D7
19	[e]	65	[E]	45	E5	05	85	C5
20	[r]	72	[R]	52	F2	12	92	D2
21	[t]	74	[T]	54	F4	14	94	D4
22	[y]	79	[Y]	59	F9	19	99	D9
23	[u]	75	[U]	55	F5	15	95	D5
24	[i]	69	[I]	49	E9	09	89	C9
25	[o]	6F	[O]	4F	EF	0F	8F	CF
26	[p]	70	[P]	50	F0	10	90	D0
27	[l]	5B	[L]	7B	DB	5B	DB	FB
28	[;]	5D	[:]	7D	DD	5D	DD	FD
29	[N]	5C	[n]	7C	DC	5C	DC	FC
30	capslock	00		00	00	00	00	00
31	[a]	61	[A]	41	E1	01	81	C1
32	[s]	73	[S]	53	F3	13	93	D3
33	[d]	64	[D]	44	E4	04	84	C4
34	[f]	66	[F]	46	E6	06	86	C6
35	[g]	67	[G]	47	E7	07	87	C7
36	[h]	68	[H]	48	E8	08	88	C8
37	[j]	6A	[J]	4A	EA	0A	8A	CA
38	[k]	6B	[K]	4B	EB	0B	8B	CB
39	[l]	6C	[L]	4C	EC	0C	8C	CC
40	[.]	3B	[,]	3A	BB	3B	BB	BA

KEYBOARD specification

41	[']	27	[']	22	A7	27	A7	A2
43	enter	0D		0D	0D	0D	0D	0D
44	shift	00		00	00	00	00	00
46	[z]	7A	[Z]	5A	FA	1A	9A	DA
47	[x]	78	[X]	58	F8	18	98	D8
48	[c]	63	[C]	43	E3	03	83	C3
49	[v]	76	[V]	56	F6	16	96	D6
50	[b]	62	[B]	42	E2	02	82	C2
51	[n]	6E	[N]	5E	EE	1E	9E	DE
52	[m]	6D	[M]	5D	ED	1D	9D	DD
53	[,]	2C	[<]	3C	AC	2C	AC	BC
54	[.]	2E	[>]	3E	AE	2E	AE	BE
55	[/]	2F	[?]	3F	AF	2F	AF	BF
57	shift	00		00	00	00	00	00
58	ctrl	00		00	00	00	00	00
60	supersh.	00		00	00	00	00	00
61	space	20		20	A0	20	A0	A0
62	super sh.	00		00	00	00	00	00
64	ctrl	00		00	00	00	00	00
75	insert	156		156	1D6	156	1D6	1D6
76	pc del.	106		157	1D7	157	1D7	1D7
79	cu left	08		11	15	02	15	15
80	home	1E		1E	1E	1E	1E	1E
81	end	155		155	1D5	155	1D5	1D5
83	cu up	0B		13	17	04	17	17
84	cu down	0A		12	16	03	16	16
85	page up	158		158	1D8	158	1D8	1D8
86	page dwn	159		159	1D9	159	1D9	1D9
89	cu right	0C		14	18	05	18	18
90	numlock			no code send				
91	home	11E	[7]	137	11E	11E	11E	1B7
92	cu left	108	[4]	134	115	102	115	1B4
93	end	165	[1]	131	1E5	1E5	1E5	1B1
95	[/]	12F	[/]	12F	1AF	12F	1AF	1AF
96	cu up	10B	[8]	138	117	104	117	1B8
97	[5]	135	[5]	135	1B5	135	1B5	1B5
98	cu down	10A	[2]	132	116	103	116	1B2
99	insert	168	[0]	130	1E8	130	1E8	1B0
100	[*]	12A	[*]	12A	1AA	12A	1AA	1AA

KEYBOARD specification

101	page up	166	[9]	139	1E6	166	1E6	1B9
102	cu right	10C	[6]	136	118	105	118	1B6
103	page dwn	133	[3]	133	1B3	133	1B3	1B3
104	pc del.	106	[.]	12E	106	106	106	1AE
105	[-]	12D	[-]	12D	1AD	12D	1AD	1AD
106	[+]	12B	[+]	12B	1AB	12B	1AB	1AB
108	enter	10D		10D	10D	10D	10D	10D
110	esc	11B	11B	11B	11B	11B		11B

112	F1	80		88	90	98	90	90
113	F2	81		89	91	99	91	91
114	F3	82		8A	92	9A	92	92
115	F4	83		8B	93	9B	93	93
116	F5	84		8C	94	9C	94	94
117	F6	85		8D	95	9D	95	95
118	F7	86		8E	96	9E	96	96
119	F8	87		8F	97	9F	97	97
120	F9	180		188	190	198	190	190

121	F10	181		189	191	199	191	191
122	F11	182		18A	192	19A	192	192
123	F12	183		18B	193	19B	193	193
124	prt sc	140		140	1C0	(*)141	1C1	1C1
125	scrolck	142		142	1C2	142	1C2	1C2
126	pause	143		143	1C3	(**)144	1C4	1C3

(*) Sys Req key.

(**) Break key.

8.4.3 Keyboard character code table 2

- Keyboard state: CAPSLOCK = irrelevant

NUMLOCK = on

NR	Normal char	word	Shift char	word	SShift word	Ctrl word	SS+Ctrl word	S+SS word
91	home	137	[7]	11E	1B7	137	1B7	19E
92	cu left	134	[4]	108	1B4	134	1B4	115
93	end	131	[1]	165	1B1	131	1B1	1E5
95	[/]	12F	[/]	12F	1AF	12F	1AF	1AF
96	cu up	138	[8]	10B	1B8	138	1B8	18B
97	[5]	135	[5]	135	1B5	135	1B5	1B5
98	cu down	132	[2]	10A	1B2	132	1B2	18A
99	insert	130	[0]	168	1B0	130	1B0	1E8
100	[*]	12A	[*]	12A	1AA	12A	1AA	1AA
101	page up	139	[9]	166	1B9	139	1B9	1E6
102	cu right	136	[6]	10C	1B6	136	1B6	118
103	page dwn	133	[3]	187	1B3	133	1B3	1E7
104	pc del.	12E	[.]	169	1AE	12E	1AE	1E9
105	[-]	12D	[-]	12D	1AD	12D	1AD	1AD
106	[+]	12B	[+]	12B	1AB	12B	1AB	1AB
108	enter	10D		10D	10D	10D	10D	10D
110	esc	11B		11B	11B	11B	11B	11B

9 Appendix A. Remarks for Joysticks

9.1 Green book specifications

- Minimum 16 directions equally spaced around the full 360 degrees are needed. (FFGB:A VII.2-17).
- A pointing device must be able to point every pixel on the screen in normal resolution, the pt_coord coordinates are in high resolution.

9.2 Digital and analog joystick

Following examples show some parameters that should be considered when developing a digital or an analog joystick.

9.2.1 Joystick DIGITAL type (3 speeds)

- It should be possible to pinpoint every pixel on the screen. So when the stick is in the first position (from center), the X and/or Y data sent should be +1 or -1. (This is the slowest speed).
- Assume we want to move the cursor over the screen in about 1 second.
 - 720 pixels in 1 second.
 - The Joystick transmits data packets at a rate of 40 per second.
 - So the value (+ or -) that should be transmitted is 720/40 or appr. \$12.
 - This could be the value for the outermost position in X or Y direction. (This is the highest speed).
- A value for the middle position could be \$09. (This is the middle speed).
- For the digital type joystick this gives following values :
 - Positive : \$01,\$09 and \$12.
 - Negative : \$FF,\$F7 and \$EE.

9.2.2 Joystick ANALOG type

- Assume again we want to move over the screen in about 1 second. So the maximum value for X and Y (in + or - direction) should be \$12.
- The center position should give \$00.
- The remaining data values (\$01 ... \$11) should be spread over the joystick's swing.
- Note : Compared with the mouse, the MSbits of the X and Y data bytes are less meaningful. But to stay as compatible as possible with the mouse, the data packet format is kept the same as the mouse.

9.3 Parameters

- data rate : the maximum possible data rate from joystick to CD-i player is 10 bits at 1200 baud or 1 byte every 8.3ms. A complete data packet takes : $3 * 8.3 = 25\text{ms}$. So the maximum data rate is 40 packets per second.
- pixels in X and Y: normal resolution = $384 * 280$.
- speed: How fast do we want to move over the screen ?

General formula : $D = NP / (SP * DR)$

D : Data value at maximum speed.

NP: Number of pixels.

SP : Cursor speed (in seconds).

DR: Data rate (in datapackets per second).

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1 Contents

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 - 2.1 Interrupts and DMA
- 3. Hardware specification
 - 3.1 Base address
 - 3.2 Floppy disk controller (DP8473)
 - 3.3 Local DMA control logic and interrupt logic
- 4. Floppy disc descriptors

2 Software Specifications

The distributed device driver for the CD-i floppy disc extension is conform to the standard OS-9 memory module format. The device driver is called by the standard OS-9 Random Block File Manager through the offset table. The driver routines are executed in system state, and the standard parameters are passed via the CPU registers :

- (a1) device descriptor module address.
- (a2) static device driver variable area address.
- (a4) process descriptor address.
- (a5) pointer to the calling process register stack.
- (a6) system global variable area address.

These driver routines are called via an entry point offset table :

DiskEnt

dc.w InitDisk	Initialize
dc.w ReadDisk	Read Sector
dc.w Write	Write Sector
dc.w GetStat	Get Status
dc.w PutStat	Put Status
dc.w Term	Terminate device
dc.w 0	Trap handler (0=none)

2.1 Interrupts and DMA

The device driver is fully interrupt driven, furthermore the read sector, write sector and write track operations are using local DMA on board to improve data transfer speed. Note that the local DMA does not use any DMA channel of the main board and consequently has no influence on its global performances.

During DMA cycle, the main processor always has access to the sector buffer RAM. It is the responsibility of the device driver to solve bus arbitration. During the execution phase of the controller operations, the device driver goes sleeping, access to the sector buffer is then reserved only for DMA transfer. At end of the execution phase as the DMA transfer is done, the controller sends an interrupt. Then the interrupt service routine wakes up the device driver. The sector buffer becomes free for the device driver.

The 68070 CPU always gets the highest priority to access the sector buffer during interrupt vector acquisition time. We assume in the design that the sector buffer address range is only owned by the device driver module. Other modules are not allowed to access the sector buffer.

3 Hardware Specifications

The CD-i floppy extension contains the decode logic, the floppy disk controller, the local DMA control logic, the sector buffer and the interrupt vector acquisition logic. The phase detector/amplifier and the VCO of the DP8473 FDC make up the phase-lock loop (PLL). They adjust the clock used during data read to keep it in phase with the data signal. No trimmer or adjustment are required.

The extension is supporting the following transfer data rates:

250 kb/s MFM (125 kb/s FM).

500 kb/s MFM (250 kb/s FM).

Standard 3" 1/2 floppy disk drives with the following characteristics are supported.

Double density 1.0MBytes capacity drives with 80 tracks per side at 300 rpm.

Quad density 2.0MBytes capacity drives with 80 tracks per side at 300 rpm.

3.1 Base address

The on board address decode logic selects a block of 512K byte out of a range of 16M byte. The base address is hardwired as 280000h.

The mapping of 512K block :

280001h	4K byte sector buffer
281FFFh	IRQ vector
282001h	8 register floppy disc controller
282011h	reset address counter
282021h	reserved
28FFFFh	

3.2 Floppy Disk Controller (DP8473)

This controller is a full featured floppy disk controller that is software compatible with the uPD765A, but also includes many additional hardware and software enhancements. The controller incorporates a precision analog data separator that includes a self trimming delay line and VCO. The chip contains 40 mA floppy drive control registers. Low power CMOS, with power down mode. The floppy disk controller contains in internal address decode logic.

The registers are mapped as follows :

Absolute address	R/W	Register description
282001	x	not used
282003	x	not used
282005	w	drive control register
282007	x	not used
282009	r	main status register
28200B	r/w	data register
28200D	x	not used
28200F	w	data rate register
28200F	r	disc change bit

For further information on the DP8473 see:

"Mass storage handbook of National Semiconductor 1989".

3.3 Local DMA control logic and interrupt logic

A local Sector buffer of 4K byte is used for higher throughput towards the host. The Highest address byte is used for the interrupt vector. Via the address of the reset address counter, one can change direction of the DMA path.

A read on reset address counter will select DMA from sector buffer to Host.

A Write on reset address counter will select DMA from Floppy disk controller to sector buffer.

4 Floppy disc descriptors

The next list gives an overview of all 3.5 inch floppy disc formats that are used in the Philips professional CD-i player. On the next pages, the descriptor parameter settings for these different formats are shown in detail.

d0	Manager Capacity	RBF $256 \times 16 \times 80 \times 2 = 653.824$ bytes
d0h	Manager Capacity	RBF $256 \times 28 \times 77 \times 2 = 1.100.800$ bytes
d0hd	Manager Capacity	RBF $256 \times 34 \times 80 \times 2 = 1.388.032$ bytes
d0tb	Manager Capacity	RBF $256 \times 16 \times 80 \times 2 = 655.360$ bytes
d0uv	Manager Capacity	RBF $256 \times 16 \times 79 \times 2 = 647.168$ bytes
pcd0	Manager Capacity	PCF $512 \times 9 \times 80 \times 2 = 737.280$ bytes
pcd0h	Manager Capacity	PCF $512 \times 18 \times 80 \times 2 = 1.474.560$ bytes

Note : All floppy disc descriptors are updated.

- additional field: PD_Rate, PD_MaxCnt
- multisector i/o is enabled: PD_Cntl

Capacity : $PD_SSize * PD_SCT * PD_CYL * PD_SID$

Table 1:

PARAMS	d0	d0h	d0hd	d0tb	d0uv
PD_DTP	1	1	1	1	1
PD_DRV	0	0	0	0	0
PD_STP	3	3	3	3	3
PD_TYP	\$00	\$00	\$00	\$20	\$20
PD_DN	\$03	\$05	\$05	\$03	\$03
PD_CYL	80	77	80	80	79
PD_SID	2	2	2	2	2
PD_VFY	0	0	0	0	0
PD_SCT	16	28	34	16	16
PD_TOS	10	16	16	16	16
PD_SAS	8	8	8	8	8
PD_ILV	2	2	2	2	2
PD_TFM	1	1	1	1	1
PD_TOffs	0	0	0	0	1
PD_SOffs	0	0	0	1	1
PD_SSize	256	256	256	256	256
PD_Cntl	\$03 or \$02	\$03 or \$02	\$03 or \$02	\$03 or \$02	\$03 or \$02
PD_Trys	7	7	7	7	7
PD_LUN	0	0	0	0	0
PD_WPC	0	0	0	0	0
PD_RWR	0	0	0	0	0
PD_Park	0	0	0	0	0
PD_LSNOffs	0	0	0	0	0
PD_TotCyls	80	77	80	80	80
PD_CtrlID	0	0	0	0	0
PD_Rate	\$10	\$10	\$10	\$10	\$10
PD_ScsiOpt	0	0	0	0	0
PD_MaxCnt	\$1fff	\$1fff	\$1fff	\$1ff	\$1fff

Table 2:

PARAMS	pcd0	pcd0h	Remarks
PD_DTP	1	1	
PD_DRV	0	0	
PD_STP	3	3	
PD_TYP	\$20	\$20	
PD_DN	\$03	\$05	
PD_CYL	80	80	
PD_SID	2	2	
PD_VFY	1	1	no verify!
PD_SCT	9	18	
PD_TOS	9	18	
PD_SAS	4	4	
PD_ILV	2	2	
PD_TFM	0	0	
PD_TOffs	0	0	
PD_SOffs	1	1	
PD_SSize	512	512	
PD_Cntl	\$03	\$03	format inhibit!
PD_Trys	0	0	
PD_LUN	0	0	
PD_WPC	0	0	
PD_RWR	0	0	
PD_Park	0	0	
PD_LSNOffs	0	0	
PD_TotCyls	80	80	
PD_CtrlID	0	0	
PD_Rate	\$10	\$10	
PD_ScsiOpt	0	0	
PD_MaxCnt	\$1fff	\$1fff	

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2 Introduction

This CD-i extension contains following features:

- SCSI interface.
- Parallel printer port.
- Ethernet port.
- RAM.
- ROM.

3 SCSI Interface

3.1 SCSI Hardware

The SCSI interface is using the DP5380 SCSI CMOS controller, for more details on the specification of this chip refer to the Mass Storage Handbook from National Semiconductor.

To achieve the highest possible throughput, the DP5380 is working in DMA mode with interrupt-driven signalling. The DMA transfer occurs between the SCSI controller and a local RAM buffer. The local RAM buffer is accessible by the CPU at any time. It is the responsibility of the device driver to resolve bus arbitration between the CPU and the SCSI controller.

Physical layer complies to the ANSI X3T9.2 standard (type single-ended).

Pin Assignment

Pin Number	Signal
2	-DB0
4	-DB1
6	-DB2
8	-DB3
10	-DB4
12	-DB5
14	-DB6
16	-DB7
18	-DBP
20	GROUND
22	GROUND
24	GROUND
26	TERMPWR

Pin Number	Signal
28	GROUND
30	GROUND
32	-ATN
34	GROUND
36	-BSY
38	-ACK
40	-RST
42	-MSG
44	-SEL
46	-C/D
48	-REQ
50	-I/O

All odd pins except pin 25 are connected to ground. Pin 25 is left open.

Connector is 50 pins type 57LE serie with a metallic shell (Amphenol).

3.2 SCSI Software

3.2.1 RBF and PCF Device Descriptors

The device driver can access disks via the RBF on three drives. Drive selection is determined by the following parameters:

- PD_CtrlrID : SCSI-Controller ID.
- PD_LUN : Logical Unit Number of the SCSI drive (always zero).

- PD_DRV : Drive number is 0, 1 or 2 . In the supplied descriptor, the drive number is equal to the controller ID and this is 0, 1, 2 for /h0, /h1, /h2 respectively.
For each of the three supported drive numbers, three distinct RBF device descriptors are supplied, e.g. for drive 0:

- “/h0fmt” should be used for formatting the disk, if a logical sector size of 512 is required. It may also be used for other disk access (e.g. read) but will give an error (000:249) if the disk has not been formatted with a logical sector size of 512.
- “/h0fix” should be used for formatting the disk, if a logical sector size of 256 is required. It may also be used for other disk access (e.g. read) but will give an error (000:249) if the disk has not been formatted with a logical sector size of 256.
- “/h0” should be used for all normal disk accesses. This will cause the driver to determine the appropriate logical sector size by reading the first sector, and to proceed accordingly.

Drive dependant option parameters are the following:

- PD_Cntl : b0: must be 0 in order for formatting to be performed with this descriptor.
b1: set multi-sector I/O capability.
b2: stable ID. This bit is set for all supplied descriptors. This accelerates disk access, but means that you must reset the system if you e.g. change the cartridge in a “syquest” drive.
b3: auto size enable. This bit is set in the supplied descriptors. This ensures that the “get capacity” SCSI command is used to ascertain the disk capacity and that the parameters PD_CYL etc. are not used.

The above bit definitions are all standard OS-9 definitions as laid down by Micro-ware. The following definition is an enhancement by Philips, in order to offer greater compatibility with disks written on (or to be read on) older CDI 605T systems or other CD-i systems:

- b15: if set means that a logical sector size of 256 should be used irrespective of the physical sector size. This only has effect when formatting a device. At all other times, the driver automatically adapts to the actual logical sector size of the medium.

- PD_TotCyls : Total number of cylinders on device.
PD_CYL : Number of cylinders per disk.
PD_SID : Number of sides.
PD_SCT : Sectors/tracks.
PD_T0S : Sectors/track on track00, must be equal to PD_SCT.
PD_SAS : Segment allocation size (typically 8).

PD_ILV : Interleave factor.

Following parameters are optional and default values are assumed:

PD_STP : Step rate (default disk value used).
 PD_DNS : Density (default disk value used).
 PD_VFY : No disk verification.
 PD_TFM : Not used.
 PD_TOffs : Drive dependant, usually zero.
 PD_SOffs : drive dependant, usually 0.
 PD_SSize : Most hard discs have a sector size of 512 bytes. These disks can be handled in two ways:
 - Let OS-9 use a 512 byte logical sector size. This is the most efficient.
 - Let OS-9 use a 256 byte logical size and do the necessary deblocking in the driver. This is less efficient, but guarantees compatibility with old releases.
 PD_Trys : Not Used.
 PD_WPC : Not Used.
 PD_RWR : Not Used.
 PD_Park : Not Used (almost all SCSI hard disks are autopark).
 PD_LSNOffs : Drive dependant, usually zero. A non-zero value usually indicates a partitioned disk.

The following PCF device descriptors are also provided:

- “/pch1” allows a PC-DOS formatted SCSI disk connected as SCSI-id 1 to be accessed.
- “/pch2” allows a PC-DOS formatted SCSI disk connected as SCSI-id 2 to be accessed.

Both the above descriptors relate to non-partitioned disks. Descriptors for partitioned disks can be derived from them using the “partdgen” utility which is included in the Extension EPROM.

For convenience a special descriptor “/pch1_” is included, which accesses the first DOS partition of a SCSI disk attached to SCSI-id 1, assuming that partition starts 32 physical sectors from the start of the physical disk. This corresponds to the default partitioning of a Syquest cartridge which has been prepared using the “SQPREP” utility.

The “DevCon” field of all RBF and PCF SCSI device descriptors indicate that the low-level device “scsi0” is to be employed to perform the actual SCSI commands.

3.2.2 SBF Device Descriptor

The descriptor “/mt0” can be used to access an Exabyte tape drive. The SCSI id of the drive should be set to 4.

The descriptor “/mtv” can be used to access an Archive Viper tape drive. The SCSI id of the drive should be set to 3.

The port address of each of the above descriptors is conventionally set equal to the SCSI id. This is never used as a port address, but is merely an artifice to ensure that a distinct static storage area is allocated for each device.

The DevCon field of the device descriptor indicates to the driver which sort of device is being used, and which device is to be used for the SCSI transfer, i.e. “scsi0/EXABYTE” or “scsi0/viper”.

The following Path descriptor fields are drive dependant:

- PD_CtrlrID : SCSI-Controller ID (to be set to match the physical drive setting, i.e. 3 for Viper, 4 for Exabyte).
- PD_LUN : Logical Unit Number of the SCSI drive (always zero).
- PD_DRV : Drive number is always zero since the drive has its own static storage.

- PD_BlkSize : This logical block size is also used as the physical block size in the case of an Exabyte drive. In other words, this value divided by 512 is the default blocking factor. Range 1 to 64 may be used, in the case of the “Viper” drive, however, a physical block size of 512 is always used, and the driver must convert between physical and logical blocks.

3.2.3 SCSI Commands

The various high level device drivers use the following SCSI commands:

- Test Unit Ready Command Operation Code 00h.
- Re-zero unit Command Operation Code 01h.
- Re-tension tape Command Operation Code 02h.
- Request sense Command Operation Code 03h.
- Format all blocks Command Operation Code 04h.
- Read block(s) Command Operation Code 08h; only for tape devices.
- Write block(s) Command Operation Code 0Ah; only for tape devices.
- Write filemark Command Operation Code 10h.
- Space the tape Command Operation Code 11h.
- Find out about device Command Operation Code 12h; only used by the “inquire” command.
- Mode select Command Operation Code 15h.
- Erase tape Command Operation Code 19h.

- Mode sense Command Operation Code 1Ah.
- Load/unload unit Command Operation Code 1Bh; only used for Vipor tape drive.
- Prevent/allow medium removal Command Operation Code 1eh; only used for TEAC tape drive.
- Read capacity Command Operation Code 25h.
- Read block(s) extended Command Operation Code 28h.
- Write block(s) extended Command Operation Code 2Ah.

These SCSI Commands are executed by calling the low-level SCSI driver “scsi605”.

3.2.4 UBF (Low level) Device Descriptor

The actual SCSI transfers are not handled directly by the RBF disk driver or by the SBF tape driver. Instead, these are handled by calling a low-level SCSI device driver. This driver is associated with the file manager UBF (=Undefined Block File manager) and with the descriptor “/scsi0”. The options fields of this descriptor determine the following:

- byte 0-3 should not be changed.
- byte 4 = SCSI id to be assumed by the CD-I 605T itself, default 5.
- byte 5, bit 0 = 1 if the SCSI bus should be reset when “/scsi0” gets attached (e.g during the very first hard disk access), default 0. Other bits reserved.
- byte 6, bit 0 = 1 if DMA channel 2 of the 68070 is to be used to transfer data between the SCSI RAM and other RAM. In this case bit 1 = 1 if these transfers may be overlapped with the actual SCSI bus transfers. These options should not be used, i.e. byte 6 should be zero (default value).
- byte 7 is reserved.
- bytes 8 through 11 contain a long word indicating the maximum DMA transfer that may be undertaken. The default value is 4Kb. Larger SCSI transfers are broken up into several DMA processes using the intermediate end-of-process facility of the SCSI controller. Exceptionally, if the high level driver indicates that the addressed target can't operate in this way (this is the case with the Exabyte tape drive), the whole transfer is done as one process if possible, irrespective of the value in this field.

The rest of the options fields should not be modified.

4 Memory Extension (DRAM and EPROM)

4.1 Hardware

The memory extension is able to support 4 banks of dynamic RAM with either (256 kB * 4) or (1 MB * 4) DRAMS.

A bank of 4 * (256kB * 4) has a capacity of 512 kBytes.

A bank of 4 * (1MB * 4) has a capacity of 2 MBytes.

Memory refresh, RAS and CAS timing generation are performed by two PALs.
All signals are synchronized by a basic clock (12MHz).

Performances:

- zero wait state for read/write cycles, assuming a 68000 processor running at 16MHz and 100ns DRAM with the key timings Trac 100ns, Tcac 30ns.
- worst case forced refresh introduces 4 wait states.

Configuration:

A set of two jumpers (factory settings) is used to configure the memory following the type of plugged DRAM.

	256			1024
A21	o	o	o	A19
gnd	o	o	o	A20
gnd	o	o	o	A21

All three jumpers in 256 position for 256k x4 chips, position 1024 for 1024kb x4 chips.

4.2 Software

The memory space is completely free for OS-9 user's application.

4.3 EPROM bank

Two JEDEC sockets are reserved for the system software (Device drivers, Internet module, etc.).

5 Centronics port

5.1 Hardware

Centronics IBM compatible interface built with standard TTL logic.

3 registers accessible by the CPU as follows:

* Data Output Register: (write only)

- 8 bits data output to the interface.

* Control Output Register: (read/write)

- b0 STROBE* : An active pulse, minimum of 500ns, clocks data into the device.
Valid data must be present for a minimum of 500ns before and after the strobe pulse.
- b1 AUTOFD* : When set to 1, auto line feed.
- b2 INIT* : When cleared starts the device (50 microseconds pulse, minimum).
- b3 SELECT* : When set to 1, selects the device.
- b4 IRQEN* : When set to 1, allows an interrupt to occur when ACK* changes from active to inactive.

* Status Register: (read only)

- b3 ERROR* : When cleared indicates the device has encountered an error.
- b4 SEL : When set to 1 indicates the device is selected.
- b5 PE : When set to 1 indicates the device has detected the end of the paper.
- b6 ACK* : Current state of the acknowledge signal.
A zero means the device has received the character and is ready to accept the next.
- b7 BUSY : A zero indicates the device cannot accept data.

Registers addressing :

Data Output Register : X E0000h

Control Output Register : X E8000h

Status Register : X F0000h

Where X stands for the extension base address nibble. (see extension memory mapping).

Connector

The port has a 25-pin, D-shell female connector with the following assignment:

Pin	I/O	Signal
1	I/O	STROBE*
2	O	D0
3	O	D1
4	O	D2
5	O	D3
6	O	D4
7	O	D5
8	O	D6
9	O	D7
10	I	ACK*
11	I	BUSY
12	I	PE
13	I	SEL

Pin	I/O	Signal
14	I/O	AUTOFD*
15	I	ERROR*
16	I/O	INIT*
17	I/O	SEL
18	N/A	GROUND
19	N/A	GROUND
20	N/A	GROUND
21	N/A	GROUND
22	N/A	GROUND
23	N/A	GROUND
24	N/A	GROUND
25	N/A	GROUND

5.2 Software

The proposed implementation is a device driver for a parallel printer. The device driver is interrupt driven, interrupt is generated by the ACK* signal on the Centronics interface.

The device driver complies to the SCF device driver structure of OS-9. That means, for the printer device a device descriptor has to be used. Device descriptor name is /p.

6 Local Area Network Interface

6.1 Hardware

The LAN interface is built with the following parts:

- AMD7990 Local Area Network Controller for Ethernet (LANCE), this communication controller manages descriptor rings and data buffer in memory. It shares memory with the CPU.
- AMD7992 Serial Interface Adapter, Manchester Encoder/Decoder compatible with IEEE-802.3, Cheapernet specifications.
- AMD7996 IEEE-802.3/Ethernet/Cheapernet transceiver.
- The LANCE performs DMA master cycles on its local RAM buffer, two 32kB x8 chips. During a DMA cycle the addresses coming from the LANCE are latched by 2 x 74HCT573 to the local RAM buffer address bus.
- During CPU cycle addresses are passed through a pair of 74HCT245 connected to the local RAM buffer address bus. A pair of 74HCT245 connects the CPU data bus to the local RAM buffer data bus.
- Bus arbitration between CPU access and LANCE access is done with control signals coming from a GAL16V8.

Interface:

Only the Cheapernet interface is implemented (BNC).

Network address:

- A DIP-switch bank with 8 positions is used to select the last byte of the network address. This DIP-switch value can be read by the CPU and passed to the Ethernet device driver.
- The 5 upper bytes of the network address will be given by a module in memory.

Three upper bytes are the manufacturer's ID (Philips).

The bytes 4 and 5 are the specific network system ID.

The last byte being the workstation address.

6.2 Software

The complete specifications are closely related to the Internet software package from Microware. The proposed implementation follows the Microware's implementation for the AMD7990 LANCE, please refer to the following document :

Using OS-9/INTERNET from Microware.

OS-9 Internet Portpak version 1.4 or higher (dependant on release).

7 Extension Memory Mapping

The base address of the Extension is programmable by jumper setting on the board.

X+0 00000 h	512 kBytes EPROM
X+0 80000 h	LANCE Registers Ethernet Buffer
X+0 A0000 h	SCSI DMA
X+0 C0000 h	DP5280 SCSI Controller Registers 32 Kb SRAM
X+0 E0000 h	Centronics Registers and Node Address
X+1 00000 h	4 MBytes Dynamic Memory

Note : Base address X = 5

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PHILIPS IMS Interactive Media Systems	605T TOOLS MENU
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2 Introduction

The start up screen of the CDI 605T has an extra icon. This icon, located in the upper right corner, displays the amount of extended memory (ROM/RAM in Mbyte) the system has selected. Selecting this icon calls up the 605T tools menu. The CDI 605T is specially designed as an authoring player. Application developers can now test their designs with special functions. Using these functions one is able to simulate some worst case and/or base case conditions for the evaluation of CD-i applications (by means of an external cd-drive (Emulator) or the internal cd-drive).

The following pages will describe these extra functionalities in more detail.

Note : This specification only represents Software release 1.3 for the CDI 605T.

3 Emulator

When using the emulator switch possibilities, an internal drive or an external connected drive can be controlled. Switching is possible as follows :

- using *the remote switch* (cinch) at the rear of the player.
- select in *the Emulator area* the CONTROL INT/EXT icon (605T tools menu).
- using *the CD-drive selector* at the rear of the player.

The following tables indicate when an internal drive or an external drive is selected :

Remote switch (cinch)	0	1	tri state
605T menu			
internal	int	ext	int
external	int	ext	ext

fig.1 : used when player shell is active

Remote switch (cinch)	0	1	tri state
CD-drive selector			
in	int	ext	int
out	int	ext	ext

fig.2 : used when player shell is inactive

int = internal drive selected
ext = external drive selected

Remark : When 'Autostart cdi' is ON and the emulator status is put to external, the Player Shell will put 'Autostart cdi' back to OFF. This to prevent a deadlock situation when no external drive is connected.

4 Additional Memory

Selecting in the Additional Memory area the INC or DEC icon, one can increase resp. decrease the amount of extended RAM and ROM in steps of 0.5 Megabyte. The sum of the ROM and RAM memory is less or equal than the maximum available extension memory. The ROM/RAM values are written in the `player_shell_settings.prf` file.

Before executing the RAM/ROM search, the NVR is searched for a '`player_shell_settings.prf`' file to get the amount of additional memory (RAM and ROM) wanted by the user. If the NVR is not initialized or the `player_shell_settings.prf` file is not found (the search happens through the complete NVR), no settings are present.

When the NVR is initialized and a `player_shell_settings.prf` file is found, the number of additional ROM blocks (blocks of 512Kb) wanted by the user is fetched out of this file. Next the RAM/ROM search will be performed.

After the kernel has started, all blocks of the ROM list (including additional ROM blocks) will be searched by the kernel for modules. The launcher will take care that the amount of RAM blocks is conform the `Ram_block` parameter in the `player_shell_settings.prf` file.

Note:

- When the additional RAM memory is set to 0.0, the launcher will start up the CD-i application in exact the base case environment. In base case, following RAM is available : for plane A : 480K + \$800 bytes, for plane B : 480K.
- In base case only the X descriptors are available. Scf and nil are not available in base case.
- For a short startup time select ROM = 0.0 .
- When the Extension module is removed, the base case requirement of at least 2 x 480k is not guaranteed anymore.
- Standard CD-i systems normally have slidly more RAM then the 2 x 480k. To avoid problem during application/title development it is advised to use only colored memory.

Application note: By using the utility '`mload`' (one can load a module in a specified place in memory with the '`-l`' option), modules can be loaded into the extension area identified as ROM. Now the modules in this area will be identified as memory modules and as long as the player is powered, the extension area defined as ROM will keep its contents (even when the player is reset). Hereby we can easily debug modules or just run special driver/applications.

5 Seek Delay

5.1 Definitions

Following terms are used throughout the document :

- *ss_seek call* : ss_seek conform the green book done by the application (a direct physical seek).
- *ss_seek+delay* : ss_seek call performed by the driver to simulate the worst possible cd-drive. (this is the normal ss_seek + an additional delay).
- *lseek/ implied seek* : an ss_play by the application is usually preceded by a lseek. This lseek updates the pointer to the desired seek position but doesn't perform the physical seek. This physical seek is performed executing the ss_play command and is indicated as implied seek.
- *seek_delay* : the additional delay.

5.2 Seek delay

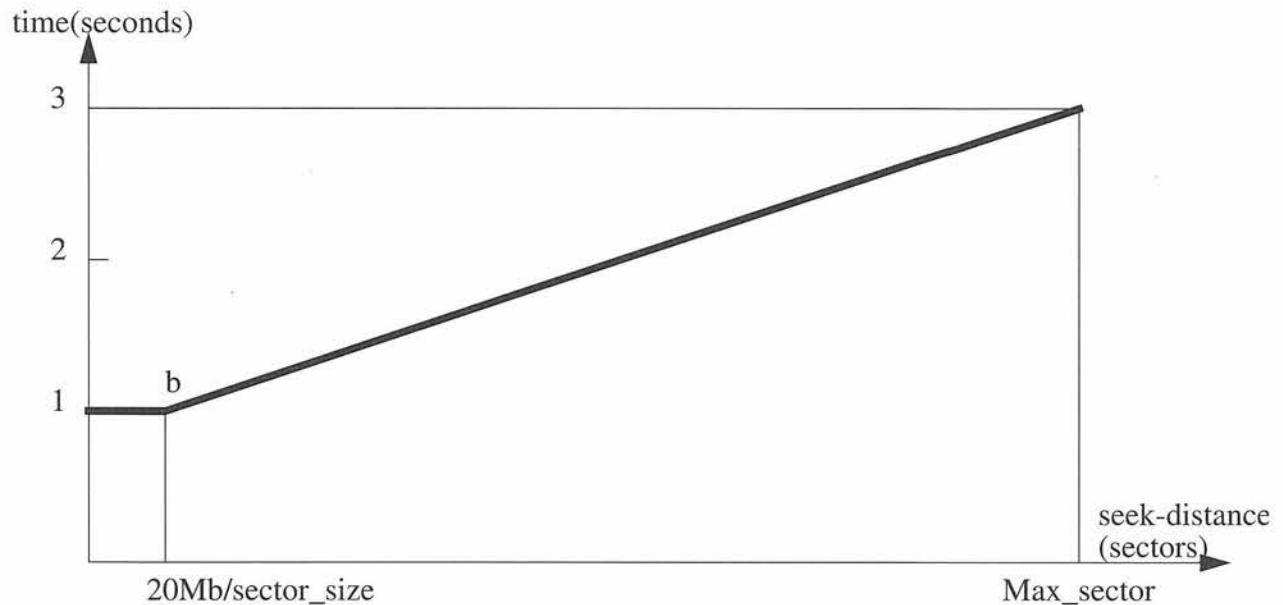
Purpose :

If seek_delay is ON in the 605T tools menu, the worst possible cd-drive is created which is still according to the green book (FFGB chapter VIII-24).

Function :

Transform a player dependent seek curve (which depends on cd device, temperature, ...) to a fixed seek curve (according to the worst possible cd device allowed by the green book) by software delays.

Hereby each seek which is performed by the application will perform following curve :



The parameters are explained.

$a = (3 - 1) / [(Max_sect) - (20Mb/sect_size)]$. (direction coeff. of the line)

$Max_sec = 72 \text{ min } 02 \text{ sec} = 4322 \text{ sec} * 75 \text{ sect/sec}$.

$sect_size = 2048$.

$b = 1$.

$t = a.s + b$ if $s > 20 \text{ Mb/sector_size}$ else $t = 1$. (time in seconds)

Note : seek-times are only valid if the disc is already spinning (not in stop mode).

One has to count also with latency of the disc. Worst case this could give a deviation of 400 ms. (300 ms latency at the outmost range of the disc + 5*13 ms for synchronization (jump 5 sectors before the wanted sector) + 30 ms of delay time for the servo processor software). Because this seek delay is software driven and measured, a deviation of 2 ticks (20 ms) can arise due to the measurements and a deviation of 5 ms on the calculations (division).

This means an `ss_seek` call will apply to the above defined curve. Now the *implied seek* of an `ss_play`, `ss_raw` or `I$Read` will be preceded by an `ss_seek+delay` call.

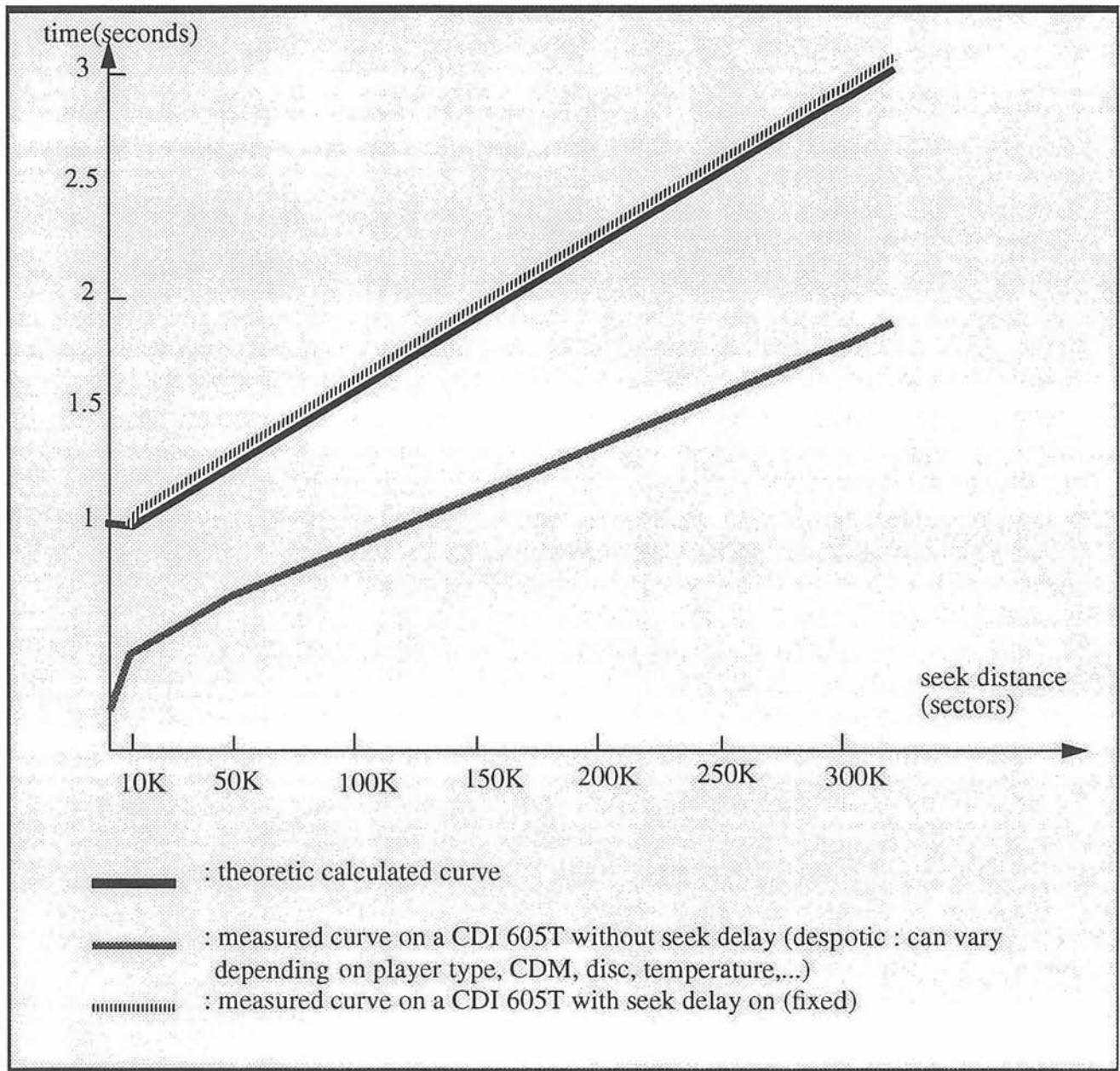
Conclusion:

- an ss_seek call will give a ASYend after the ss_seek is ended (the ss_seek is ended when the physical seek is ended and the additional seek delay is finished).
- an ss_play call will perform first an ss_seek+delay and starts then the ss_play really. The application gets back control when waiting for the ss_seek+delay to be ended and the play to be started up.
- an ss_raw and I\$Read call will perform first an ss_seek+delay and start the ss_raw or I\$Read only if the ss_seek+delay is ended. The application gains control after the ss_raw or I\$Read is performed. While performing the ss_seek+delay, a sleep is done until the ss_seek+delay is ended and then the ss_raw or I\$Read command is given.

Notes:

- on the ss_cdda no seek_delay is added. This is a red function and we've skipped this function against the three other green functions which will normally be used more in applications.
- if seek_delay is set off, one has to take into account a few more tests in the cdapdriv which must be done to conclude if a seek_delay is desired or not. The time to perform these tests should be neglected by the applications.
- an ss_play performs first an ss_seek+delay and then the implied seek. This implied seek time will be very small because the ss_seek+delay performed a physical seek to the desired position. So the implied seek will be only of a few sectors and could be neglected.
- when an application gives an ss_seek call followed by an ss_play call, it must pay attention to the two seek_delays (according to the curve). The first from the ss_seek and the second from the ss_play, the second one will normally be 1 second (distance < 20 Mb/sector_size). The implied seek could again be neglected.

Following diagram is measured with *ss_seek* calls:



Notes :

- The above measurements are performed using a test program which jumps a certain number of times between two sectors and calculates the average.
- Due to the start of the calculation of the direction coeff. (not 1 second anymore) a small nod arises around the 10K of the measured curve with seek delay on.

- The measured curve with seek delay on is a little bit higher than the theoretical calculated curve due to the deviations described above (calculations, ...). Latency times are not shown here, they do not occur using the `ss_seek` call. They can occur by an `ss_play`, `ss_raw` or `I$Read`.

6 Video settings

6.1 Purpose:

Testing video and audio initialization procedures of CD-i applications in order to make them compatible with other types of CD-i players.

6.2 Function:

Randomize the video settings of the video IC before starting a CD-i application.

6.3 Method:

- On a CDI 605T the player shell will always overwrite the hardware registers of the video IC before starting a CD-i application. These registers are overwritten by executing mostly an FCT/LCT which contains Control Program Instructions (the other instructions will be video functions e.g. interlace).
- These instructions will be saved into the binary file `'/nvr/vid_regs.prf'` (See later for the proper structure of this file). This file always represents the video settings at the start of the last ran CD-i application. However, if Control Program Instructions are not executed correctly, a message will be sent to the standard output device and the `'invalid'`-field in the video settings file will be set. The tool `'rdvid'`, which is a ROM-module, creates the opportunity of looking at the content of any saved video settings file or even change it.
- The execution of LCT/FCT instructions before starting a CD-i application depends on the video settings mode in the 605T tool menu. By manipulating `'Previous'` and `'Randomize'` 3 different situations can be obtained:

PREV	RAND		
OFF	ON	->	Control Program Instructions will be randomized before starting a CD-i application, saved into the video settings file and executed.
ON	OFF	->	The Control Program Instructions will be read from the video settings file and will be executed.

OFF OFF -> The Control Program Instructions from the FCT/LCT used by the player shell will be saved into the video settings file and executed (default option).

- Randomizing a Control Program Instruction means that we take the lowest legal value for that instruction and 'OR' this with a 24 bit randomized value. In this case a valid instruction is always obtained.

6.4 Structure of the video settings file :

The video settings file are located in '/nvr/vid_regs.prf'

Offset	Plane	Action	NOP	(1)
0	-	invalid-field	-	(3)
4	A	Load display parameters	N	
8	A	Signal when this line reached	Y/N	(2)
12	A	Set border color	N	
16	A	Select image coding methods	N	
20	A	Load transparency control info	N	
24	A	Load plane order	N	
28	A	Load transparent color	N	
32	A	Load mask color	N	
36	A	Load DYUV start value	N	
40	A	Load backdrop color	N	
44	A	Load mosaic pixel hold factor	N	
48	A	Load image contribution factor	N	
52	A	Set CLUT bank 0	N	
56	A	Load CLUT color / QHY level 0 - 7	N	
88	A	Load CLUT color 8 - 63	N	
312	A	Set CLUT bank 1	N	
316	A	Load CLUT color / QHY level 0 - 7	N	
348	A	Load CLUT color 8 - 63	N	
572	A	Load matte register 0 - 7	N	
604	B	Load display parameters	N	
608	B	Signal when this line reached	Y/N	(2)
612	B	Set border color	N	
616	B	Load transparent color	N	
620	B	Load mask color	N	
624	B	Load DYUV start value	N	
628	B	Load mosaic pixel hold	N	
632	B	Load image contribution factor	N	
636	B	Set CLUT/QHY bank 2	N	
640	B	Load CLUT color / QHY level 0 - 7	N	
672	B	Load CLUT color 8 - 63	N	

896	B	Set CLUT/QHY bank 3	N	
900	B	Load CLUT color / QHY level 0 - 7	N	
932	B	Load CLUT color 8 - 63	N	
1156	B	Load matte register 0 - 7	N	
1188	-	Set display interlace mode	-	(3)
1190	-	Set compatibility mode	-	(3)
1192	-	Audio attenuation	N	
1196	-	Tickperiod for PAL	-	(3)
1198	-	Tickperiod for NTSC	-	(3)

(1) When an instruction is not used in the FCT/LCT then the NOP (No Operation) instruction is written into the video settings file. In the table above for every instruction is indicated if the NOP instruction is used instead of the instruction itself.

N : instruction

Y : NOP instruction

(2) N/ Y : A randomized boolean will decide if the signal-instruction or the NOP instruction is used.

(3) The following data fields are no Control Program Instructions:

- invalid-field:

0 -> this file is valid, so all instructions may be interpreted, else this file is invalid, because executing FCT/LCT failed or writing an instruction to the FCT/LCT failed.

- Set display interlace mode: FFGB. VII-177.

0 -> interlace OFF.

1 -> interlace ON.

- Set compatibility mode: FFGB. VII-176.

0 -> compatibility OFF.

1 -> compatibility ON.

- Audio attenuation: FFGB. VII-69.

This field contains the attenuation value (32 bits) itself.

- Tickperiod for PAL/NTSC : FFGB VIII-13.

When randomizing the video settings, the player shell installs the default, shortest or longest clock tick period within the 0.2% tolerance. The following values are defined:

PAL	NTSC	Meaning	Tickperiod PAL	Tickperiod NTSC
----	----	-----	-----	-----
0x61d	0x628	longest tick period	10.0160 ms	10.0163 ms
0x61a	0x625	default tick period	9.99680 ms	9.99730 ms
0x618	0x623	shortest tick period	9.98400 ms	9.98450 ms

The PAL and NTSC values are the ones of the reload register from the 68070.

The player shell will always use the tick period value of the actual video mode he is running in.

Note: All Control Program Instructions used by the player shell are located in the FCT's. The LCT's are empty.

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1 Contents

Use of Launcher

\$ launcher -c <application name> startup application from disk

\$ launcher -f <application name> startup application from floppy

Reserved for additional info such as application notes.

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